$\Rightarrow (E_1' \cup E_2')' = E_1 \cap E_2$ is measurable.

Topic # 42

Theorem: For any $\phi \neq A \subset R$ and $\{E_i\}_{i=1}^{n}$ be a family of pairwise disjoint measurable sets, then

$$m^*[A \cap (\bigcup_{i=1}^{n} E_i)] = \sum_{i=1}^{n} m^*(A \cap E_i) \rightarrow (A) \quad n \in N$$

proof: We will prove this by mathematical induction. For $n=1$, result is self-evident.

Suppose (A) is true for $n = k-1$;

$$\therefore (A) \Rightarrow m^*[A \cap (\bigcup_{i=1}^{k-1} E_i)] = \sum_{i=1}^{k-1} m^*(A \cap E_i) \rightarrow (B)$$

Consider: $A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k = A \cap [(\bigcup_{i=1}^{k} E_i) \cap E_k]$

$$= A \cap [(E_1 \cup E_2 \cup \cdots \cup E_k) \cap E_k]$$

$$A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k = A \cap [\underbrace{(E_1 \cap E_k)}_{=\phi} \cup \underbrace{(E_2 \cap E_k)}_{=\phi} \cup \cdots \cup (E_k \cap E_k)]$$ ✓

$\because E_i \cap E_j = \phi \quad \forall i \neq j$

$$\Rightarrow A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k = A \cap E_k \rightarrow (1)$$

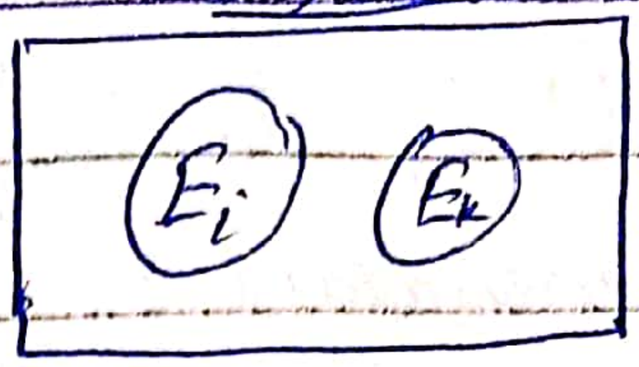
Now

$$A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k' = A \cap [(\bigcup_{i=1}^{k} E_i) \cap E_k']$$

$$= A \cap [(E_1 \cup E_2 \cup \cdots \cup E_k) \cap E_k']$$

$$= A \cap [(E_1 \cap E_k') \cup (E_2 \cap E_k') \cup \cdots \cup \underbrace{(E_k \cap E_k')}_{\phi}]$$

Disjoint



$E_i \subseteq E_k'$

$\Rightarrow E_i \cap E_k' = E_i$

$\Rightarrow A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k' = A \cap (E_1 \cup E_2 \cup \cdots \cup E_{k-1})$

$\Rightarrow A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k' = A \cap (\bigcup_{i=1}^{k-1} E_i) \longrightarrow (2)$

Taking "$A \cap (\bigcup_{i=1}^{k} E_i)$" as test set and using Lebesgue measurability of "E_k":

$$m^*\left[A \cap (\bigcup_{i=1}^{k} E_i)\right] = m^*\left[A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k\right] + m^*\left[A \cap (\bigcup_{i=1}^{k} E_i) \cap E_k'\right] \quad \text{By defi}$$

$$m^*\left[A \cap (\bigcup_{i=1}^{k} E_i)\right] = m^*(A \cap E_k) + m^*\left[A \cap (\bigcup_{i=1}^{k-1} E_i)\right]$$

$$= m^*(A \cap E_k) + \sum_{i=1}^{k-1} m^*(A \cap E_i)$$

$$m^*\left[A \cap (\bigcup_{i=1}^{k} E_i)\right] = \sum_{i=1}^{k} m^*(A \cap E_i)$$

$\therefore$ Truth of $n = k-1$ also implies its truth for $n = k$. This proves the given theorem.

Topic #43

Theorem: The class "M" of all measurable sets is a σ-algebra.

$\phi = X$; $\mathcal{A} \subseteq P(X)$ (subsets of "X")

(i) $\forall A \in \mathcal{A} \Rightarrow A' \in \mathcal{A}$

(ii) $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$

(iii) $\{E_i\}_{i=1}^{\infty} \rightarrow \bigcup_{i=1}^{\infty} E_i \in \mathcal{A}$

proof: (i) If "E" is measurable

$\rightarrow$ "E'" is also measurable

$\rightarrow E \in$ "M"

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(ii) If E_1, E_2 are measurable $\Rightarrow$ $E_1 \cup E_2$ is also measurable.

$\Rightarrow E_1 \cup E_2 \in M$

(iii) Let $\{E_i\}_{i=1}^{\infty}$ be the sequence of measurable sets in "M", Then we will show

$$\bigcup_{i=1}^{\infty} E_i = E \in M$$

Constracting: $F_1 = E_1$, $F_2 = E_2 - E_1 \Rightarrow F_2 \cap F_1 = \phi$

$F_3 = E_3 - (E_1 \cup E_2) \Rightarrow F_3 \cap F_2 = \phi$

$\vdots$ $\vdots$

$F_n = E_n - (E_1 \cup E_2 \cup \cdots \cup E_{n-1}) = E_n - \left(\bigcup_{i=1}^{n-1} E_i\right)$

$\Rightarrow \{F_i\}_{i=1}^{\infty}$ is a sequence of pairwise disjoint measurable sets.

$$\Rightarrow \bigcup_{i=1}^{\infty} F_i = \bigcup_{i=1}^{\infty} E_i \longrightarrow (1)$$

Let $H_n = \bigcup_{i=1}^{n} F_i$ - i.e. a measurable as well being finite union of measurable sets.

$$\Rightarrow H_n = \bigcup_{i=1}^{n} F_i \subseteq \bigcup_{i=1}^{\infty} F_i = \bigcup_{i=1}^{\infty} E_i = E$$

$\Rightarrow H_n' \supseteq E' \Rightarrow \forall A \subseteq R, A \cap H_n' \supseteq A \cap E'$

$\Rightarrow m^*(A \cap H_n') \geq m^*(A \cap E') \longrightarrow (2)$

| $A \subseteq B$
 $A' \supseteq B'$

$\because$ "H_n" is measurable and taking $\phi \neq A \subseteq R$ as test set:

$\Rightarrow m^*(A) = m^*(A \cap H_n) + m^*(A \cap H_n')$ * By defi:

$$\Rightarrow m^*(A) \geq m^*(A \cap H_n) + m^*(A \cap E')$$

$$m^*(A) \geq m^*\left[A \cap \left(\bigcup_{i=1}^{n} F_i\right)\right] + m^*(A \cap E')$$

$$m^*(A) \geq \sum_{i=1}^{n} m^*(A \cap F_i) + m^*(A \cap E')$$

$\because$ L.H.S is independent of "n", so for $n \to \infty$

$$\therefore m^*(A) \geq \sum_{i=1}^{\infty} m^*(A \cap F_i) + m^*(A \cap E')$$

$$\Rightarrow m^*(A) \geq m^*\left[A \cap \left(\bigcup_{i=1}^{\infty} F_i\right)\right] + m^*(A \cap E')$$

$$\Rightarrow m^*(A) \geq m^*\left[A \cap \left(\bigcup_{i=1}^{\infty} E_i\right)\right] + m^*(A \cap E')$$

$$\Rightarrow m^*(A) \geq m^*(A \cap E) + m^*(A \cap E')$$

$\Rightarrow$ "E" is also measurable.

$\Rightarrow E \in M$

$\Rightarrow$ "M" is a σ-algebra.

Topic #44

Theorem: If "F" is measurable set and $m^*(F \Delta G) = 0$, then "G" is measurable set.

proof: $G - F \subseteq (F \Delta G)$

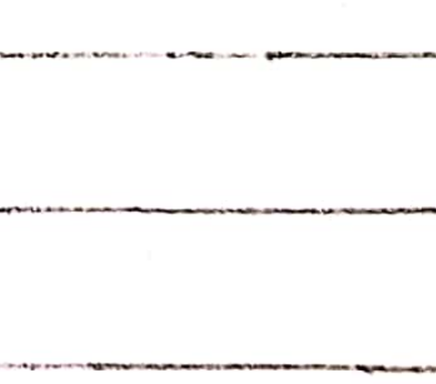
$\Rightarrow m^*(G-F) \leq m^*(F \Delta G) = 0$

$\Rightarrow m^*(G-F) = 0$

$\because m^*$ is non-negative.

$\Rightarrow (G-F)$ is measurable.

Similarly, $(F-G)$ is also measurable.



$F \Delta G = (G-F) \cup (F-G)$

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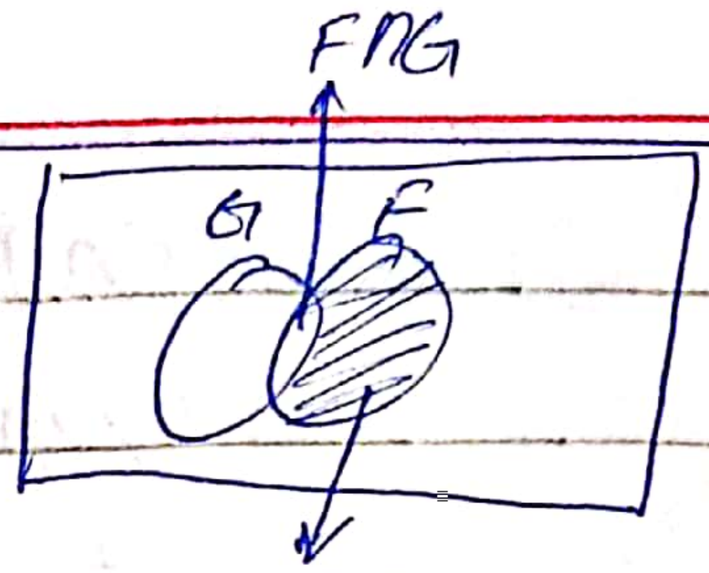
$\because$ $F \cap G = F \cap (F - G)'$

is measurable being

intersection of two measurable sets.

$\therefore$ $G = (F \cap G) \cup (G - F)$ is also measurable.



Topic # 45

G-Delta and F-Sigma Sets:

G_δ-Set

A $\phi \neq A \subseteq \mathbb{R}$ is said to be G_δ-set if it can be written as countable intersection of open sets.

For Example:

$$\bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n}\right) = (a, b)$$

(a, b) is trivially G_δ-set.

Note: G_δ-set is measurable being the countable intersection of measurable sets.

F_σ-set

A $\phi \neq A \subseteq \mathbb{R}$ is said to be F_σ-set if it can be written as countable union of closed sets.

For Example:

$$(a, b) = \bigcup_{n=1}^{\infty} \left[a + \frac{1}{n}, b - \frac{1}{n}\right]$$

(a, b) is also F_σ-set.

Note: F_σ-set is measurable being the countable ~~intersection~~ union of measurable sets.

Note: $G_\delta' = F_\sigma$ and $F_\sigma' = G_\delta$

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Topic #46

Theorem: Given any $\phi \neq A \subseteq \mathbb{R}$ and $\varepsilon > 0$, there is a G_δ set "G" with $A \subseteq G$ and $m^*(A) = m^*(G)$.

Proof: $\because \forall\ A \subseteq \mathbb{R}$ and for any $\varepsilon > 0$, there exist an open set say "U" such that $A \subseteq U$ and $m^*(U) < m^*(A) + \varepsilon$ $\longrightarrow (1)$

In particular for $\varepsilon = \frac{1}{n}$, let "$U_n$" an open set, such that $A \subseteq U_n$

$\Rightarrow m^*(A) \leq m^*(U_n) \longrightarrow (2)$

and $m^*(U_n) < m^*(A) + \frac{1}{n} \longrightarrow (3)$

Since $A \subseteq U_n \Rightarrow A \subseteq \bigcap_{n=1}^{\infty} U_n = G_\delta$ (say) $\longrightarrow (4)$

i.e. G_δ-set being intersection of measurable sets. (open sets)

$\Rightarrow m^*(G_\delta) = m^*\left(\bigcap_{n=1}^{\infty} U_n\right) \leq m^*(U_n)$

$\Rightarrow m^*(G_\delta) < m^*(A) + \frac{1}{n}$ (From (3))

As for $n \to \infty \Rightarrow \varepsilon = \frac{1}{n} \to 0$

$\therefore m^*(G_\delta) \leq m^*(A) \longrightarrow (5)$

From (4) $\Rightarrow A \subseteq G_\delta \Rightarrow m^*(A) \leq m^*(G_\delta) \longrightarrow (6)$

& From (5) and (6) $\Rightarrow m^*(A) = m^*(G_\delta)$

$A \subseteq B$

$\Rightarrow m^*(A) \leq m^*(B)$

$A \subseteq U$

$\Rightarrow m^*(A) \leq m^*(U)$

$m^*(A) + \varepsilon > m^*(U)$

$\Rightarrow m^*(U) < m^*(A) + \varepsilon$

where $\varepsilon > 0$

$A \cap B \subseteq A$

$m^*(A \cap B) \leq m^*(A)$

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MTH644
Measure Theory

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Topic # 47

Theorem: Interval (a,∞) is measurable $\forall\ a\in R$.

Proof:

Taking test set say $A\subseteq R$ and we only need to prove:

$\because\ m^*(A)=m^*(A\cap U)+m^*(A\cap U')$

$\because\ m^*(a,b)=l(a,b)=b-a$

$$m^*(A)\geq m^*[A\cap(a,\infty)]+m^*[A\cap(-\infty,a]] \longrightarrow (1)$$

Let $A_\alpha = A\cap(a,\infty)$ and $A_\beta = A\cap(-\infty,a]$

$\therefore (1)\Longrightarrow m^*(A)\geq m^*(A_\alpha)+m^*(A_\beta) \longrightarrow (2)$

Case I: $m^*(A)=+\infty$, then (2) is obviously.

Case II: if $m^*(A)<\infty$ and for $A\subseteq R\ \exists$ a seq. $\{I_n'\}_{n=1}^{\infty}$ such that $A\subseteq \bigcup_{n=1}^{\infty} I_n' \longrightarrow (A)$

$\forall\ \varepsilon>0$, Then there exists another such sequence $\{I_n\}$ such that

$(A)\Rightarrow m^*(A)\leq m^*\left(\bigcup_{n=1}^{\infty} I_n\right)$ "

$A\subseteq \bigcup_{n=1}^{\infty} I_n$ and $m^*(A)+\varepsilon \geq m^*\left(\bigcup_{n=1}^{\infty} I_n'\right) \longrightarrow (a)$

$\because$ on R; $m^*(I_n)=l(I_n)$

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$\therefore m^*\left(\bigcup_{n=1}^{\infty} I_n\right) = \sum_{n=1}^{\infty} l(I_n)$ → not disjoint

$\longrightarrow$ (b)

(a) $\Rightarrow \sum_{n=1}^{\infty} l(I_n) < m^*(A) + \epsilon$ $\longrightarrow$ (3)

Let $M_n = I_n \cap (a, \infty)$ and $N_n = I_n \cap (-\infty, a]$

$M_n \cap N_n = \{I_n \cap (a,\infty)\} \cap \{I_n \cap (-\infty, a]\}$

$= I_n \cap \{(a,\infty) \cap (-\infty, a]\}$

$M_n \cap N_n = I_n \cap \phi = \phi$

Now $M_n \cup N_n = \{I_n \cap (a,\infty)\} \cup \{I_n \cap (-\infty, a]\}$

$= I_n \cap \{(a,\infty) \cup (-\infty, a]\}$

$= I_n \cap R$

$M_n \cup N_n = I_n$

$\Rightarrow l(I_n) = l(M_n) + l(N_n)$ $\longrightarrow$ (4)

$\because A_\alpha = A \cap (a,\infty)$ and $A \subseteq \bigcup_{n=1}^{\infty} I_n$

$\Rightarrow A_\alpha \subseteq \left(\bigcup_{n=1}^{\infty} I_n\right) \cap (a,\infty)$

$\Rightarrow A_\alpha \subseteq (I_1 \cup I_2 \cup \dots) \cap (a,\infty)$

$\Rightarrow A_\alpha \subseteq \{I_1 \cap (a,\infty)\} \cup \{I_2 \cap (a,\infty)\} \cup \dots$

$\Rightarrow A_\alpha \subseteq \bigcup_{n=1}^{\infty} \{I_n \cap (a,\infty)\}$

$\Rightarrow A_\alpha \subseteq \bigcup_{n=1}^{\infty} M_n$

$\Rightarrow m(A_\alpha) \leq m^*\left(\bigcup_{n=1}^{\infty} M_n\right)$ $\longrightarrow$ (c)

Similarly $A_\beta \subseteq \bigcup_{n=1}^{\infty} N_n$

$\Rightarrow m(A_\beta) \leq m^*\left(\bigcup_{n=1}^{\infty} N_n\right)$ $\longrightarrow$ (d)

Adding (c) and (d)

$m(A_\alpha) + m(A_\beta) \leq m^*\left(\bigcup_{n=1}^{\infty} M_n\right) + m^*\left(\bigcup_{n=1}^{\infty} N_n\right)$

$\leq \sum_{n=1}^{\infty} m^*(M_n) + \sum_{n=1}^{\infty} m^*(N_n)$

$\because M_n \cap N_n = \phi \quad \forall n \in \mathbb{N}$

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$$m^*(A_\alpha) + m^*(A_\beta) \leq \sum_{n=1}^{\infty} l(M_n) + \sum_{n=1}^{\infty} l(N_n)$$

$$\leq \sum_{n=1}^{\infty} (l(M_n) + l(N_n))$$

$$m^*(A_\alpha) + m^*(A_\beta) \leq \sum_{n=1}^{\infty} l(I_n) \quad \text{From (4)}$$

$$\Rightarrow m^*(A_\alpha) + m^*(A_\beta) < m^*(A) + \varepsilon \quad \text{From (3)}$$

$\because \varepsilon \to 0$ arbitrarily;

$$\therefore m^*(A_\alpha) + m^*(A_\beta) \leq m^*(A)$$

or $m^*(A) \geq m^*(A_\alpha) + m^*(A_\beta)$

Hence proved; (a, ∞) is measurable.

Topic #48

Theorem: Open interval $(a,b) \subset \mathbb{R}$ is measurable.

proof: $\because$ (a, ∞) is measurable

$\Rightarrow (a,\infty)^c = (-\infty, a]$ is also measurable.

And $(-\infty, b - \frac{1}{n}]$ is also measurable $\forall\ n \in \mathbb{N}$ as $n \to \infty$

and $(-\infty, b) = \bigcup_{n=1}^{\infty} (-\infty, b - \frac{1}{n}]$ is also measurable

because being countable union of measurable set

$\Rightarrow (a,\infty) \cap (-\infty, b) = (a,b)$ i.e also measurable.

being intersection of two measurable set

Open set in $\mathbb{R}$:

Any open set in $\mathbb{R}$ say U can be written as countable union of disjoint open intervals

i.e $U = \bigcup_{n=1}^{\infty} \{I_n\}$ and each open intervals is

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measurable and countable union of measurable set is measurable.

Topic #49

Theorem: A subset $\phi \neq E \subset \mathbb{R}$ is measurable $\Leftrightarrow$ for given $\varepsilon > 0$, $\exists$ an open set $U \supseteq E$ with $m^*(U-E) < \varepsilon$.

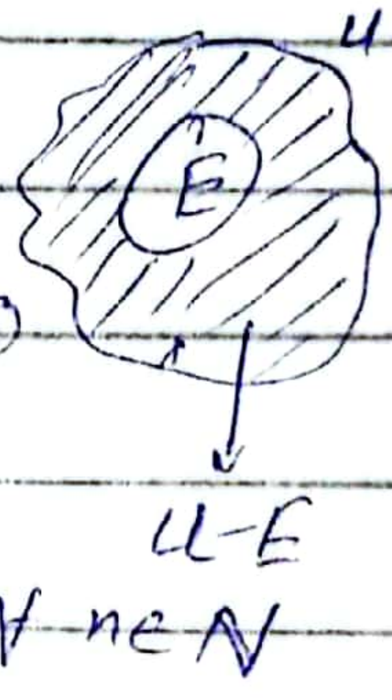
proof: Suppose E is measurable

Case I: If $m^*(E) < \infty$, $\forall\ \phi \neq E \subset \mathbb{R}$, there exists an open set U such that $E \subset U$ and for any $\varepsilon > 0$, $m^*(U) < m^*(E) + \varepsilon \longrightarrow (a)$

$E \subset U \Rightarrow U = E \cup (U-E)$

$\Rightarrow m^*(U) = m^*(E) + m^*(U-E)$

$\Rightarrow m^*(U-E) = m^*(U) - m^*(E) < \varepsilon$. from (a)



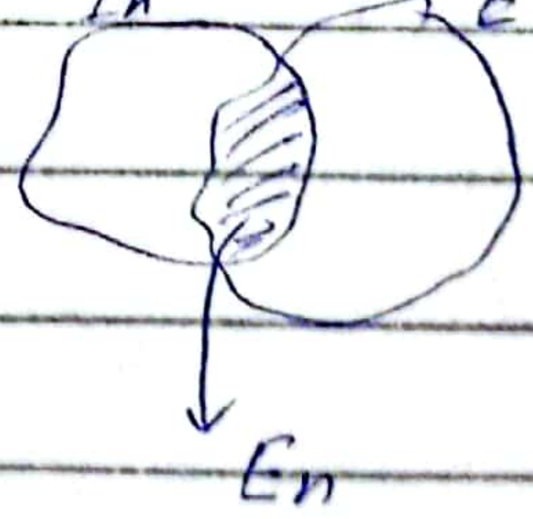
Case II: If $m^*(E) = \infty$

$\mathbb{R} = \bigcup_{n=1}^{\infty} I_n$, where $\ell(I_n) < \infty\ \forall\ n \in \mathbb{N}$

and $I_j \cap I_k = \phi$, $j \neq k$

put $E_n = E \cap I_n \Rightarrow E_n \subseteq I_n\ \forall\ n \in \mathbb{N}$

$\Rightarrow m^*(E_n) \leq m^*(I_n) = \ell(I_n) < \infty$



By case I $\forall\ E_n$, $\exists$ an open set say "V_n" such that $E_n \subseteq V_n$, for $\varepsilon > 0$

and $m^*(V_n - E_n) < \frac{\varepsilon}{2^n} \longrightarrow (b)$

Also $V = \bigcup_{n=1}^{\infty} V_n$, an open set.

$\Rightarrow \bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} (E \cap I_n) = E \cap \left(\bigcup_{n=1}^{\infty} I_n\right) = E \cap \mathbb{R} = E$

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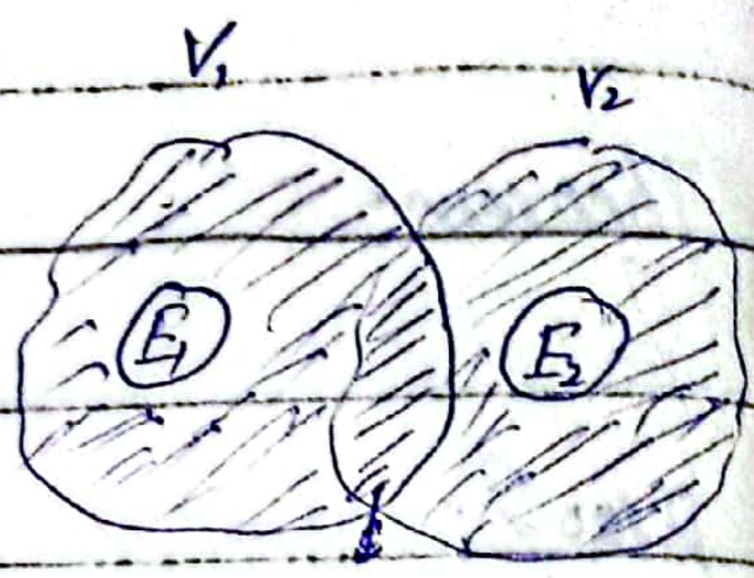
$E = \bigcup_{n=1}^{\infty} E_n \subseteq \bigcup_{n=1}^{\infty} V_n = V$ $\quad \because E_n \subseteq V_n \ \forall\, n \in \mathbb{N}$

$\Rightarrow V - E = \bigcup_{n=1}^{\infty} V_n - \bigcup_{n=1}^{\infty} E_n \subseteq \bigcup_{n=1}^{\infty} (V_n - E_n)$

$\Rightarrow m^*(V-E) \leq \sum_{n=1}^{\infty} m^*(V_n - E_n)$

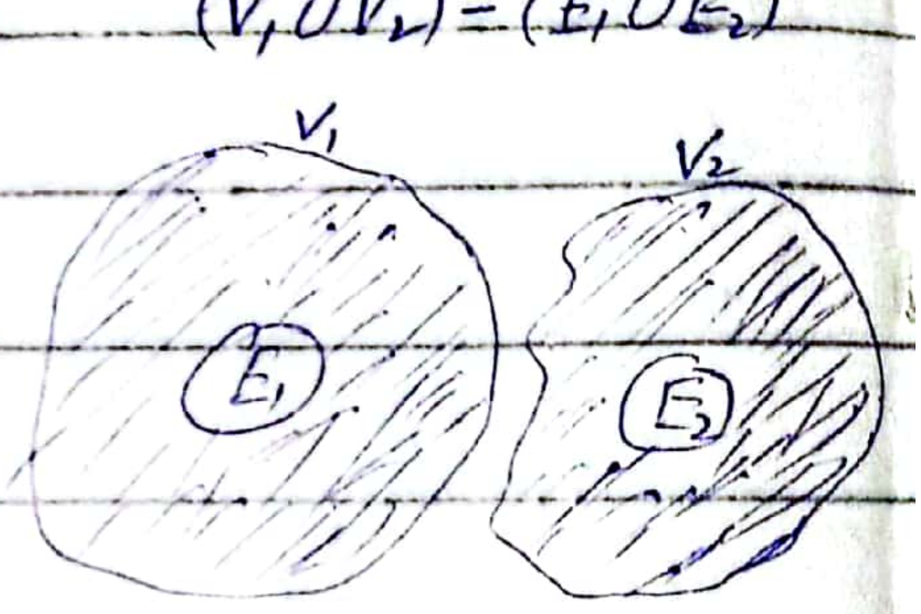


From (b) $\Rightarrow m^*(V-E) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n}$

$m^*(V-E) \leq \varepsilon\left(\sum_{n=1}^{\infty} \frac{1}{2^n}\right) = \varepsilon(1)$

$m^*(V-E) \leq \varepsilon$

$(V_1 \cup V_2) = (E_1 \cup E_2)$



$(V_1 - E_1) \cup (V_2 - E_2)$

Conversely, let $\varepsilon = \frac{1}{n}$ (say) and

$\exists$ an open set $O_n \supseteq E$ such that $E \subseteq O_n \ \forall\, n \in \mathbb{N}$

and $m^*(O_n - E) < \varepsilon = \frac{1}{n} \longrightarrow (A)$

We will prove "E" is measurable.

put $G_1 = \bigcap_{n=1}^{\infty} O_n \Rightarrow$ "G_1" is measurable being countable intersection of measurable open set.

$E \subseteq O_n \ \forall\, n$ (given)

and $E \subseteq \bigcap_{n=1}^{\infty} O_n = G_1$

$G_1 - E \subseteq O_n - E$

$\Rightarrow m^*(G_1 - E) \leq m^*(O_n - E) < \frac{1}{n} = \varepsilon$

$\because \varepsilon \to 0$ arbitrary $m^*(G_1 - E) = 0$.

$\Rightarrow G_1 - E$ is measurable $\Rightarrow (G_1 - E)'$ is also measurable.

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$\therefore E = G_1 - (G_1 - E) = G_1 \cap (G_1 - E)'$ is measurable being intersection of two measurable set.

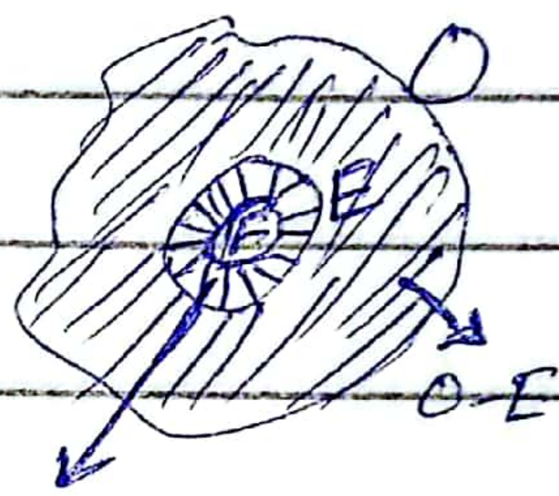
Topic #50

Theorem: For a subset "E" of R and given $\varepsilon > 0$ $\exists$ an open set $O \supseteq E$ and $m^*(O-E) < \varepsilon \Leftrightarrow \exists$ a closed set $F \subseteq E$ and $m^*(E-F) < \varepsilon$.

proof: For $\phi \neq E \subseteq R \Rightarrow E' \subseteq R$ $\exists$ for given $\varepsilon > 0$

$\exists$ an open set $O \supseteq E$ and

$m^*(O-E) < \varepsilon \longrightarrow (1)$



$O \supseteq E' \Rightarrow E'' \supseteq O' = F$ (say)

i.e. closed set

$\Rightarrow E \supseteq F \Rightarrow m^*(E-F) = m^*(E \cap F')$

$m^*(E-F) = m^*(F' \cap E)$

$m^*(E-F) = m^*(O-E') < \varepsilon$ from (1)

$\Rightarrow m^*(E-F) < \varepsilon$

Conversely for the set $E' \subseteq R$, $\exists$ a closed set "F" and $\varepsilon > 0$ such that $F \subseteq E'$ and $m^*(E'-F) < \varepsilon \longrightarrow (2)$ (given)

$F \subseteq E' \Rightarrow E = E'' \subseteq F' = O$ (open set) say.

$\Rightarrow m^*(O-E) = m^*(O \cap E') = m^*(E' \cap O)$

$\Rightarrow m^*(O-E) = m^*(E'-O') = m^*(E'-F) < \varepsilon$ from (2)

$m^*(O-E) < \varepsilon$ (Hence proved)

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Topic #52

Theorem: Let $\phi \neq E \subset R$, then the following statements are equivalent

(i) There is a G_δ set "G" with $E \subset G$, $m^*(G-E)=0$ → give

(ii) There is a F_σ set "F" with $F \subset E$, $m^*(E-F)=0$ → prove

proof: (i) $\Rightarrow$ (ii) $\because \phi \neq E \subset R \Rightarrow \phi \neq E' \subseteq R$

$\therefore$ For $\phi \neq E'$ given $E' \subseteq G$, $m^*(G-E')=0$

$\therefore \quad 0 = m^*(G-E') = m^*(G \cap (E')') = m^*(G \cap E)$

$= m^*(E \cap G) = m^*(E \cap (G')') = m^*(E-G')$

$= m^*(E-F)$, where $G_\delta = G = \bigcap_{n=1}^{\infty} G_n$

$\Rightarrow G_\delta' = G' = \left(\bigcap_{n=1}^{\infty} G_n\right)'$

$\Rightarrow G_\delta' = \bigcup_{n=1}^{\infty} G_n'$

$\Rightarrow G_\delta' = \bigcup_{n=1}^{\infty} F_n' = F = F_\sigma$

(ii) $\Rightarrow$ (i)

Conversly, again for $E' \subset R$, with $F \subset E'$

$m^*(E'-F)=0$ (given).

$\therefore \quad 0 = m^*(E'-E) = m^*(E' \cap F') = m^*(F' \cap E')$

$= m^*(F'-E) = m^*(G-E)$

where $F_\sigma' = F' = \left(\bigcap_{k=1}^{\infty} F_k\right)' = \bigcup_{k=1}^{\infty} F_k' = \bigcup_{k=1}^{\infty} G_n = G = G_\delta$

(proved)

Topic #52

Lebesgue Measure:

Let "M" be the class of measurable sets, which is σ-algebra, Then the Lebesgue measure of any set $E \subseteq M$ is defined to be its lebesgue outer measure.

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$$m^* \equiv M ; \left\{ A,\ m^*(A) = m^*(A \cap E) + m^*(A \cap E') \right.$$

Topic #53

Theorem: let $\{E_i\}_{i=1}^{n}$ be a sequence a in measurable space "M", then m is

(i) countably subadditive

(ii) finitely additive provided that $E_i \cap E_j = \phi, \forall\ i \neq j$.

(iii) countably additive if E_i's are pairwise disjoint.

(iv) "m" is monotonic.

proof: (i) All E_i's are measurable.

$\Rightarrow$ $\bigcup_{i=1}^{n} E_i$ is also measurable, $\therefore$ "M" is space of lebesgue measurable set i

$$\Rightarrow m\left(\bigcup_{i=1}^{n} E_i\right) = m^*\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} m^*(E_i) = \sum_{i=1}^{\infty} m(E_i)$$

(ii) $\forall\ \phi \neq A \subseteq R$ and for the sequence $\{E_i\}_{n=1}^{n}$ of pairwise disjoint measurable sets ;

$$m^*\left(A \cap \left(\bigcup_{i=1}^{\infty} E_i\right)\right) = \sum_{i=1}^{\infty} m^*(A \cap E_i)$$

put $A = R \Rightarrow A\left(\bigcup_{i=1}^{\infty} E_i\right) = \bigcup_{i=1}^{\infty} E_i$ $\quad A \cap E_i = R \cap E_i$

$$\Rightarrow m^*\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} m^*(E_i)$$

$$\Rightarrow m\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} m(E_i)$$

(iii) $m^*\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} m^*(E_i)$

$\Rightarrow m\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} m(E_i)$ finitly additive.

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iv Monotonicity : let $\phi \neq A, B \in M$, such that $B \subseteq A$.

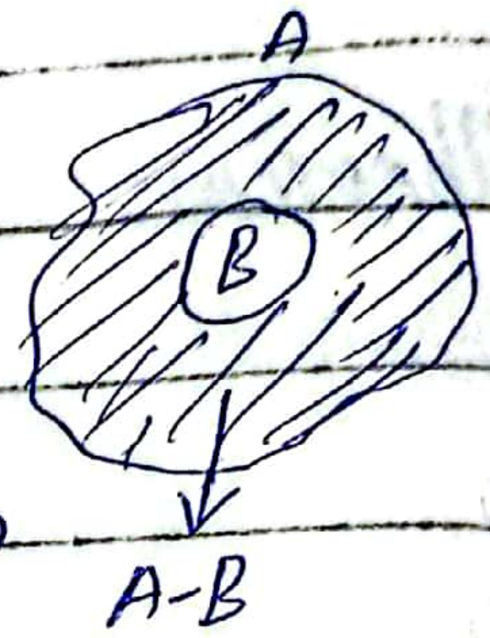
$\Rightarrow A = B \cup (A-B)$

$\therefore m^*$ is finitely additive,

$\Rightarrow m^*(A) = m^*(B) + m^*(A-B)$

$\Rightarrow m^*(A) \geq m^*(B)$ $\because m^*(A-B) \geq 0$

$\Rightarrow m^*(B) \leq m^*(A)$



$\because B \cap (A-B) \neq \phi$

Topic # 54

Theorem: Lebesgue measure "m" is translation invariant.

proof: $\because$ Given that say $\phi \neq E \in M$ is measurable.

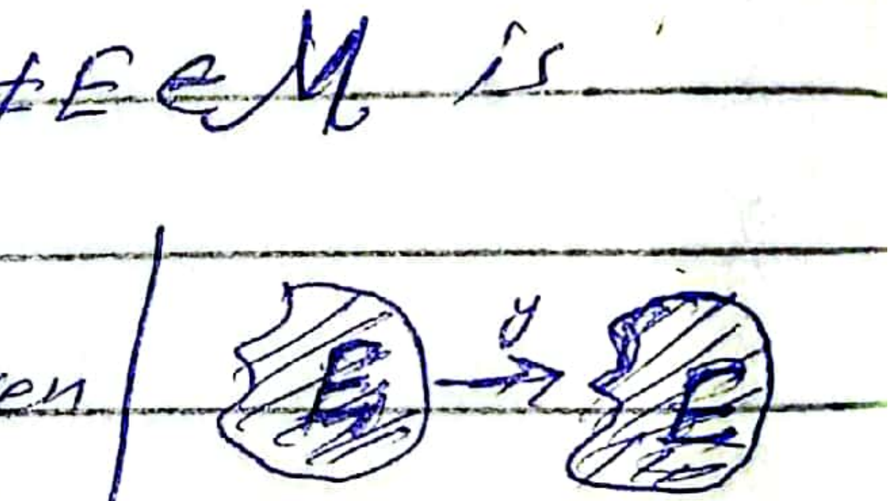
$\because$ "E" is measurable $\Leftrightarrow$ for given $\varepsilon > 0$, $\exists$ an open set $U \supseteq E$, such that $m^*(U-E) < \varepsilon$

"U" is open set in $\mathbb{R} \Rightarrow$ "U+y" is also open in $\mathbb{R}$, $\forall\ y \in \mathbb{R}$

if $E \subseteq U \Rightarrow E+y \subseteq U+y$

$\Rightarrow (U+y)-(E+y) = (U-E)+y$

$\Rightarrow m^*[(U+y)-(E+y)] = m^*[(U-E)+y]$



$\because$ Lebesgue outer measure is translation invariant.

$\Rightarrow m^*(U-E) < \varepsilon$

But $m^* = m \Rightarrow m[(U+y)-(E+y)] = m(U-E) < \varepsilon$

$\Rightarrow$ "E+y" is also measurable $\Rightarrow$ "m" is translation invariant.

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Topic # 55

Theorem: Suppose E_1 and E_2 are two measurable sets, then $m(E_1 \cup E_2) + m(E_1 \cap E_2) = m(E_1) + m(E_2)$

proof: If $m(E_1) = m(E_2) = \infty$, then the result is obvious.

Case I: If $m(E_1) < \infty$, $m(E_2) < \infty$

$\therefore$ if E_1 is measurable, then for any $\phi \neq A \subseteq R$

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap E_1') \longrightarrow (1)$$

For particularly if $A = E_2$

$$(1) \Rightarrow m^*(E_2) = m^*(E_2 \cap E_1) + m^*(E_2 \cap E_1') \longrightarrow (2)$$

Similarly $E_2 \longleftrightarrow E_1$

$$m^*(E_1) = m^*(E_1 \cap E_2) + m^*(E_1 \cap E_2') \longrightarrow (3)$$

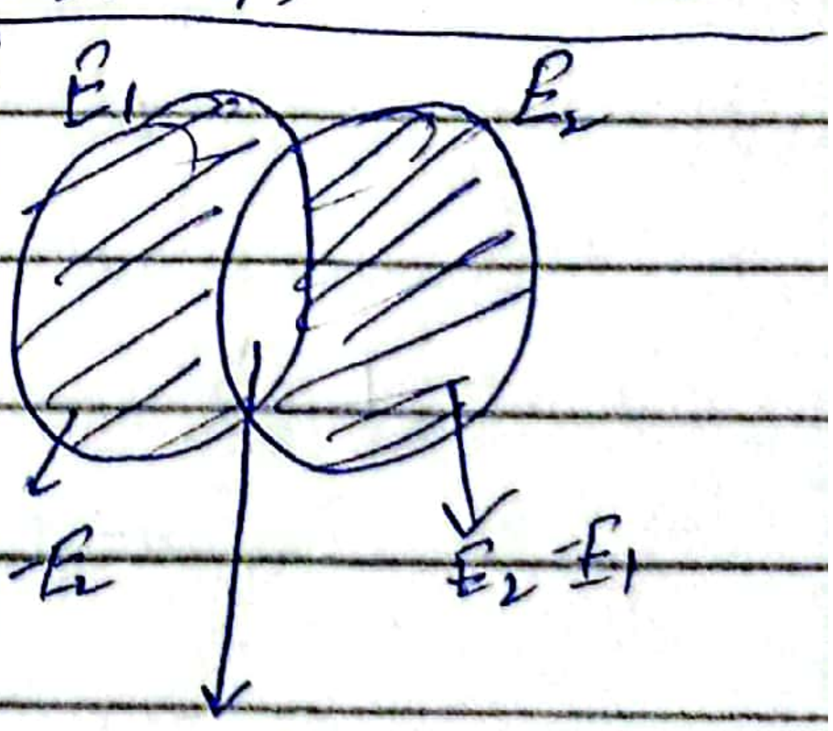
Adding (2) and (3): $(m^* \equiv m)$

$$m(E_1) + m(E_2) = m(E_1 \cap E_2) + m(E_1 \cap E_2') + m(E_2 \cap E_1) + m(E_2 \cap E_1') \longrightarrow (A)$$

$$E_1 \cup E_2 = (E_1 - E_2) \cup (E_2 - E_1) \cup E_1 \cap E_2$$

$$= (E_1 \cap E_2') \cup (E_1 \cap E_1') \cup (E_1 \cap E_2)$$

$$\Rightarrow m(E_1 \cup E_2) = m[(E_1 \cap E_2') \cup (E_2 \cap E_1') \cup (E_1 \cap E_2)]$$



"m" is finitely addable:

$$m(E_1 \cup E_2) = m(E_1 \cap E_2') + m(E_2 \cap E_1') + m(E_1 \cap E_2)$$

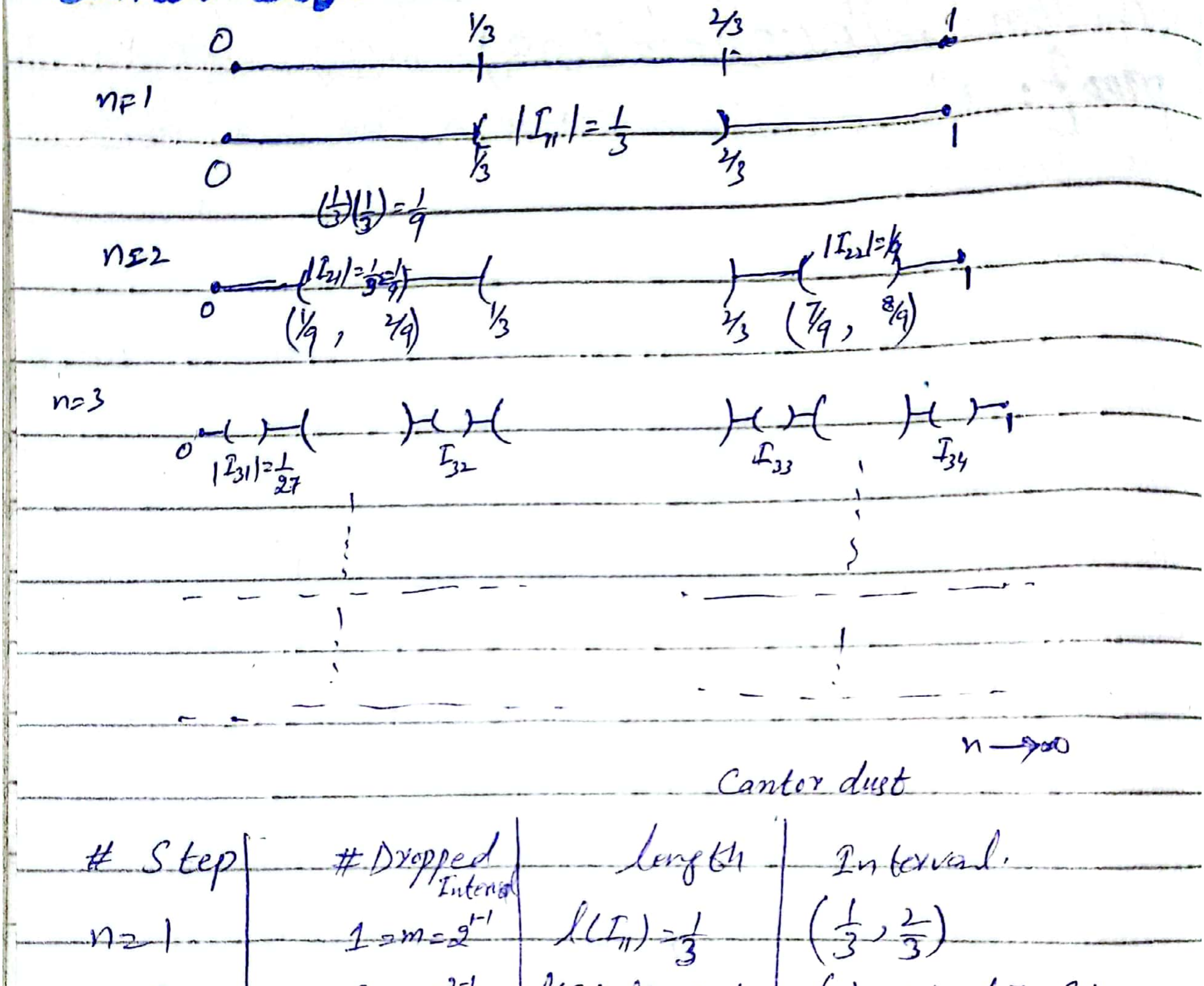
$$(A) \Rightarrow m(E_1) + m(E_2) = m(E_1 \cup E_2) + m(E_1 \cap E_2)$$

proved

Topic # 56

$R = R^2 = R^3$

Cantor Set:



Cantor dust

# Step	# Dropped Interval	length	Interval
n = 1	$1 = m = 2^{1-1}$	$l(I_{11}) = \frac{1}{3}$	$(\frac{1}{3}, \frac{2}{3})$
n = 2	$2 = m = 2^{2-1}$	$l(I_{21}) = l(I_{22}) = \frac{1}{3^2}$	$(\frac{1}{9}, \frac{2}{9}), (\frac{7}{9}, \frac{8}{9})$
n = 3	$4 = m = 2^{3-1}$	$l(I_{31}) = \dots = \frac{1}{3^3}$	
n	$m = 2^{n-1}$	$l(I_{n.}) = \dots = \frac{1}{3^n}$	

Dropped Interval = $G_1 = \underbrace{I_{11}}_{n=1}, \underbrace{I_{21}, I_{22}}_{n=2}, \underbrace{I_{31}, I_{32}, I_{33}, I_{34}, \dots}_{n=3}$

$I_{n1}, I_{n2}, \dots I_{nm}, \dots$

$n \to \infty, m = 2^{n-1}$

$\therefore G_1 = \{I_{11}\}, \{I_{21}, I_{22}\}, \{I_{31}, I_{32}, I_{33}, I_{34}\}, \dots$

$n \to \infty, m = 2^{n-1}$

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$$\Rightarrow \quad G_1 = \bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}$$

↗ universal set.

$\therefore$ Cantor Set $= C = [0,1] - G_1 = G_1'$

↘ $l = 1$

Topic #57

Measure of Cantor Set:

$\because$ "G_1" is open, being countable union of disjoint open intervals $\Rightarrow$ "G_1" is measurable.

$\because C = G_1'$ = closed and is also measurable.

$$m(G_1) = m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right) \quad \because \text{"}m\text{" is countably additive}$$

$$= \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} m(I_{nm}) = \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} l(I_{nm})$$

expanding firstly by "n"

$$= \sum_{m=1}^{2^{1-1}} l(I_{1m}) + \sum_{m=1}^{2^{2-1}} l(I_{2m}) + \sum_{m=1}^{2^{3-1}} l(I_{3m}) + \cdots\cdots$$

$$= \underbrace{l(I_{11})}_{} + \underbrace{l(I_{21}) + l(I_{22})}_{} + \underbrace{l(I_{31}) + l(I_{32}) + l(I_{33}) + l(I_{34})}_{} + \cdots\cdots$$

$$= \frac{1}{3} + \left(\frac{1}{3^2} + \frac{1}{3^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3}\right) + \cdots\cdots$$

$$m(G_1) = \frac{1}{3} + \frac{2}{3^2} + \frac{4}{3^3} + \cdots\cdots \quad \begin{cases} a = \frac{1}{3}, & r = \frac{2}{3} \end{cases}$$

$$S_\infty = \frac{a}{1-r} = \frac{1/3}{1-2/3} = 1$$

$\boxed{m(G_1) = 1}$ $\because [0,1] = G \cup C \Rightarrow m[0,1] = m(G_1) + m(C) \Rightarrow \boxed{m(C) = 0}$

↑ disjoint

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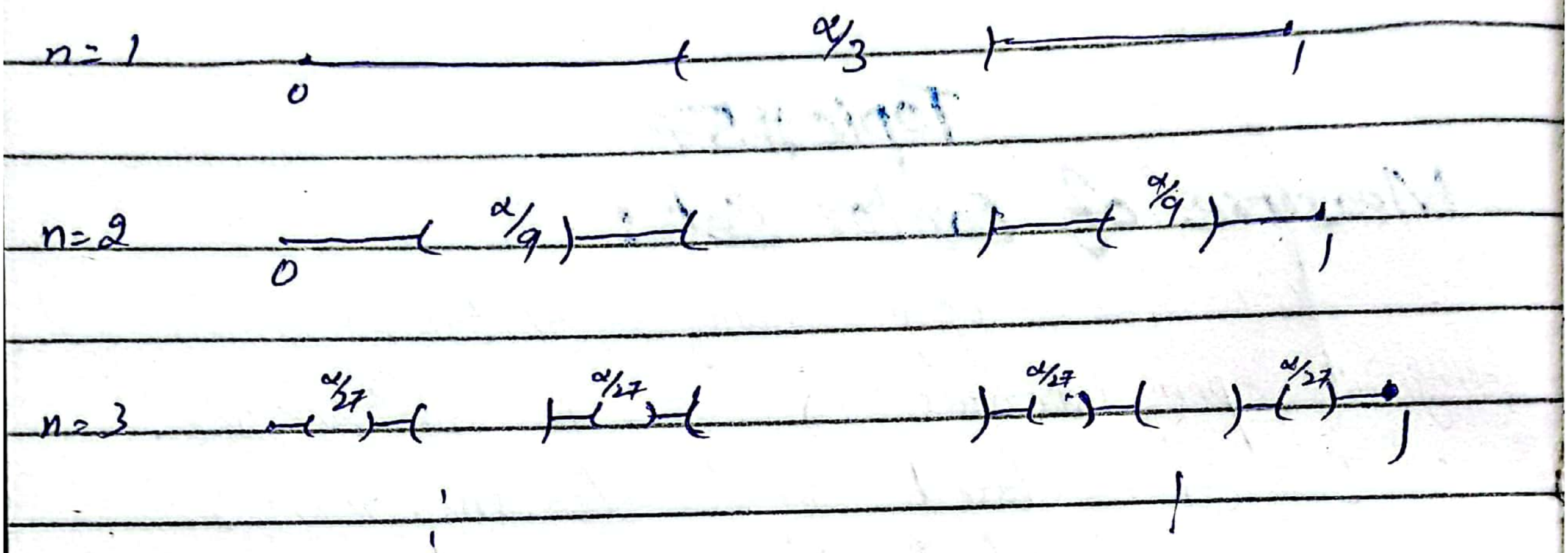
Topic #58

Generalized Cantor Set and its Measure:



$$m(G_1) = \frac{\alpha}{3} + \left(\frac{\alpha}{9} + \frac{\alpha}{9}\right) + \left(\frac{\alpha}{27} + \frac{\alpha}{27} + \frac{\alpha}{27} + \frac{\alpha}{27}\right) + \cdots$$

$$m(G_1) = \frac{\alpha}{3} + \frac{2\alpha}{3^2} + \frac{4\alpha}{3^3} + \cdots$$

$$a = \alpha/3, \quad r = 2/3$$

$$S_\infty = \frac{a}{1-r} = \frac{\alpha/3}{1-2/3} = \frac{\alpha/3}{1/3} = \alpha$$

$$\boxed{m(G_1) = \alpha}$$

$\because [0,1] = G_1 \cup C$ (disjoint)

$$\Rightarrow m[0,1] = m(G_1) + m(C)$$

$$1 = \alpha + m(C)$$

$$\Rightarrow \boxed{m(C) = 1 - \alpha}$$

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Topic # 59

Theorem: Let $\{E_n \uparrow\}_{n=1}^{\infty}$ be an increasing sequence of measurable sets. Then

$$m^*\left(\bigcup_{n=1}^{\infty} E_n\right) = m\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_n)$$

Proof: $\because$ Given that $(E_n, \uparrow)$, such that

$\Rightarrow E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots$ of measurable sets

Defining $F_1 = E_1$, $F_2 = E_2 - E_1$, $F_3 = E_3 - E_2$, $\cdots$

$F_n = E_n - E_{n-1}$, where $\{F_n\}_{n=1}^{\infty}$ is a seq. of disjoint measurable sets.

$$\Rightarrow \bigcup_{k=1}^{n} F_k = \bigcup_{k=1}^{n} E_k \Rightarrow \bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} E_k.$$

$$\therefore m^*\left(\bigcup_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty} F_k\right) \quad \text{(disjoint + measurable)}$$

$$= \sum_{k=1}^{\infty} m(F_k) = \lim_{n\to\infty} \sum_{k=1}^{\infty} m(F_k)$$

$$= \lim_{n\to\infty} m\left(\bigcup_{k=1}^{\infty} F_k\right) = \lim_{n\to\infty} m\left(\bigcup_{k=1}^{\infty} E_k\right)$$

$$= \lim_{n\to\infty} m^*(E_n) \qquad \text{(proved)}$$

Topic # 60

Theorem: Let $\{E_n \downarrow\}_{n=1}^{\infty}$ be a decreasing sequence of measurable sets such that $m(E_k) < \infty$ for some $k \in \mathbb{N}$, Then

$$m^*\left(\bigcap_{n=1}^{\infty} E_n\right) = m\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_n)$$

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proof: Let $m(E_1) < \infty$, Say (without lose of generality)

$\because \langle E_n, \downarrow \rangle \Rightarrow E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots$

$\Rightarrow \phi = E_1 - E_1 \subseteq E_1 - E_2 \subseteq E_1 - E_3 \subseteq \cdots E_1 - E_n \subseteq \cdots$

$E_1 \subseteq E_n \subseteq \cdots$

$$\Rightarrow m^*\left(\bigcup_{n=1}^{\infty}(E_1 - E_n)\right) = \lim_{n\to\infty} m^*(E_1 - E_n)$$

$$\bigcup_{n=1}^{\infty}(E_1 - E_n) = \bigcup_{n=1}^{\infty}(E_1 \cap E_n')$$

$$= (E_1 \cap E_1') \cup (E_1 \cap E_2') \cup \cdots$$

$$= E_1 \cap [E_1' \cup E_2' \cup \cdots]$$

$$= E_1 \cap [E_1 \cap E_2 \cap \cdots]'$$

$$= E_1 \cap \left(\bigcap_{n=1}^{\infty} E_n\right)'$$

$$= E_1 - \left(\bigcap_{n=1}^{\infty} E_n\right)$$

Sets Relation ↑

$$m^*\left(E_1 - \bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_1 - E_n)$$

$$m^*(E_1) - m^*\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_1) - \lim_{n\to\infty} m^*(E_n)$$

$$m^*(E_1) - m^*\left(\bigcap_{n=1}^{\infty} E_n\right) = m^*(E_1) - \lim_{n\to\infty} m^*(E_n)$$

$$\Rightarrow m^*\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_n) \quad \text{(proved)}.$$

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Topic #61

Borel Set:

e.g. $G_\delta = \bigcap_{n=1}^{\infty} U_n$; $F_\sigma = \bigcup_{n=1}^{\infty} F_n$

Definition: A set "E" is said to be a Borel set if it can be expressed as countable union or intersection of open or closed sets.

e.g. $\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x + \frac{1}{n}\right)$ → G_δ Borel sets.

Topic #62

Borel Measure Space:

Restriction of Lebesgue measure "m" to the class of Borel set "B" is said to be Borel measure space. $(\mathbb{R}, B, m)$

Theorem: Every Borel set is measurable.

proof: Every open or closed set is measurable. ⇒ Countable union or intersection of closed or open set is also measurable.

∴ Borel set is also measurable being union or intersection of closed or open sets

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Topic #63

Theorem: Countable set are Borel sets with zero measure.

Proof: Case I: Singleton:

$\forall\ a \in \mathbb{R}: \{a\} = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, a + \frac{1}{n}\right)$; $n \in \mathbb{N}$

Countable
↓
finite ⟷ infinite.
$\mathbb{Z}, \mathbb{N}, \mathbb{R}, \mathbb{Q}$ ---

$\therefore \left\{\left\{a - \frac{1}{n}, a + \frac{1}{n}\right\}, \downarrow\right\}$ is decreasing sequence.

with $m(E_n) = \frac{2}{n} < \infty$

where $E_n = \left(a - \frac{1}{n}, a + \frac{1}{n}\right)$

$$\Rightarrow m\{a\} = m\left\{\bigcap_{n=1}^{\infty} E_n\right\} = \lim_{n \to \infty} m(E_n)$$

$$= \lim_{n \to \infty} \ell(E_n)$$

$$= \lim_{n \to \infty} \left(\frac{2}{n}\right) = \frac{2}{\infty}$$

$$= 0$$

Finite Set

$$\{x_1, x_2, x_3, \dots\dots x_n\} = \bigcup_{i=1}^{n} \{x_i\}$$

$$\Rightarrow m\left(\bigcup_{i=1}^{n} \{x_i\}\right) = \sum_{i=1}^{n} m\{x_i\} = 0$$

Infinite Set:

$$\{x_1, x_2, x_3, \dots\dots\dots\} = \bigcup_{i=1}^{\infty} \{x_i\}$$

$$\Rightarrow m\left(\bigcup_{i=1}^{\infty} \{x_i\}\right) = \sum_{i=1}^{\infty} m\{x_i\} = 0$$

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Topic #64

Sum Modulo 1:

Let $x, y \in A = [0,1)$ and we define sum modulo "1" of x and y as

$$x \hat{+} y = \begin{cases} x+y & \text{if } x+y<1 \\ x+y-1 & \text{if } x+y \geq 1 \end{cases}$$

Translation Modulo 1:

Let $E \subseteq A = [0,1)$ and $y \in A$, we define a translation modulo "1" of "E" as.

$$E \hat{+} y = \{ x \hat{+} y, x \in E \}.$$

$\downarrow I$

Set of all "Sum Modulo 1"

Topic #65

Theorem: Let $E \subset A = [0,1)$ be a measurable set, Then for ~~any~~ each $y \in A$. The set $E \hat{+} y$ is measurable and $m(E) = m(E \hat{+} y)$.

proof: Suppose $E_1 = E \cap [0, 1-y)$ and

$E_2 = E \cap [1-y, 1)$

$\therefore E_1 [0, 1-y), [1-y, 0)$ are measurable

$\Rightarrow E_1$ and E_2 are also measurable being the intersection of two measurable sets.

Also $E_1 \cap E_2 = \{E \cap [1-y, 1)\} \cap \{E \cap [0, 1-y]\}$

$= E \cap \{[0, 1-y) \cap [1-y, 1)\}$

$= E \cap \phi = E$

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and $E_1 \cup E_2 = E \cup \phi = E$

$\Rightarrow m(E) = m(E_1) + m(E_2) \longrightarrow (1)$

"m" is finitely additive.

Let $y_1 \in A$, if $y_1 \in E_1$

$\Rightarrow y_1 < 1-y \Rightarrow y_1 + y < 1 \Rightarrow y_1 \hat{+} y = y_1 + y$

$E_1 + y$

$\Rightarrow E_1 \hat{+} y = E_1 + y \longrightarrow (a)$

if $y_2 \in E_2 \Rightarrow y_2 \geq 1-y \Rightarrow y_1 + y \geq 1$

$\Rightarrow y_2 \hat{+} y = y_2 + y - 1$

$\Rightarrow E_2 \hat{+} y = E_2 + (y-1) \longrightarrow (b)$

$\because$ "m" is translation invariant.

$\Rightarrow E_1 + y$ and $E_2 + (y-1)$ are measurable.

$\Rightarrow E_1 \hat{+} y$ and $E_2 \hat{+} (y-1)$ are measurable.

$\Rightarrow m(E_1 \hat{+} y) = m(E_1 + y) = m(E_1)$

Similarly

$$m(E_2 \hat{+} y) = m(E_2 + y - 1) = m(E_2)$$

$\because E_1 \cup E_2 = E$, $E_1 \cap E_2 = \phi$

$\Rightarrow E \hat{+} y = (E_1 + y) \cup (E_2 + y)$ and $(E_1 + y) \cap (E_2 + y) = \phi$

"$E \hat{+} y$" being union of two disjoint measurables sets its measurable.

$\Rightarrow m(E \hat{+} y) = m(E_1 \hat{+} y) + m(E_2 \hat{+} y)$

$= m(E_1 + y) + m(E_2 + y)$

$= m(E_1) + m(E_2) = m(E)$

$\Rightarrow$ "m" is also invariant w.r.t translation modulo "1".

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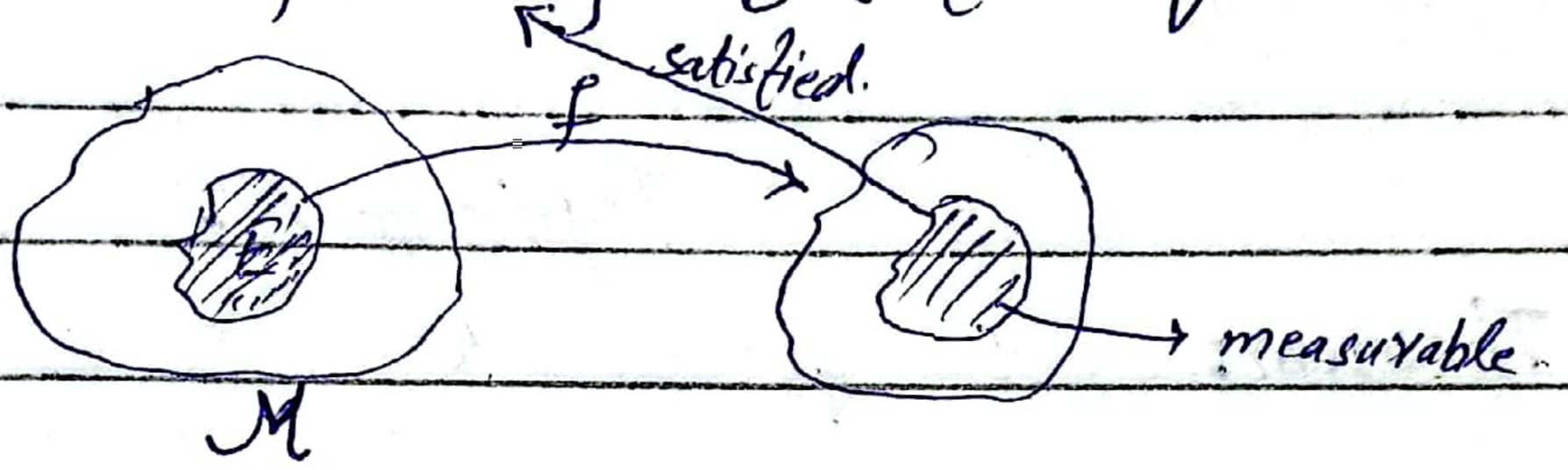
Topic # 66

Measurable Function:

Let "M" be the measurable space (σ-algebra of measurable sets). a function $f: E \to \mathbb{R}$ is Lebesgue measurable function on $\phi \neq E \in M$.

if $\{x \in E : f(x) > a\} \in M$ for every $a \in \mathbb{R}$.



For Example (1): Suppose "f" is real valued identically constant function $f(x) = \beta$, Then:

$$\{x : f(x) > a\} = \begin{cases} \mathbb{R} & \text{if } \beta > a \\ \phi \in M \end{cases}$$

$\therefore \mathbb{R}, \phi \in M$

$\Rightarrow$ "f" is measurable.

For Example (2):

$$f(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

$$\{x : f(x) > a\} = \begin{cases} \phi \\ A \neq X \\ X = A \end{cases}$$

$\phi, A, X \in M \Rightarrow$ "f" is measurable.

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Example (3): $(R, B, m) \to$ (Borel σ-algebra)

$f(x) = x$

$\Rightarrow \{x : f(x) > a\} = \begin{cases} (a, +\infty) \in B \\ \text{i.e always measurable.} \end{cases}$

$\Rightarrow$ "f" is measurable.

Topic # 67

Theorem: Let $f : E \to R \cup \{\pm\infty\}$, $\phi \neq E \in M$, Then.

(i) "f" is measurable $\Leftrightarrow$ (ii) $\{x \in E : f(x) \geq a\}$ is measurable for all $a \in R \Leftrightarrow$ (iii) $\{x \in E : f(x) < a\}$ is measurable $\forall\ a \in R \Leftrightarrow$ (iv) $\{x \in E : f(x) \leq a\}$ is measurable $\forall\ a \in R$.

Proof: (i) $\to$ (ii) "f" is measurable $\to \{x \in E : f(x) > a\} \in M$

$$\{x \in E : f(x) \geq a\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$$

Let $y \in \{x \in E : f(x) \geq a\} \Rightarrow f(y) \geq a, \forall\ a \in R$

$\Rightarrow f(y) > a - \frac{1}{n}, \forall\ n \in N$

$\Rightarrow y \in \{x \in E : f(x) > a - \frac{1}{n}\}, \forall\ n \in N$

$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$

$\{x \in E : f(x) \geq a\} \subset \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$ $\rightarrow$ (a)

Let conversely $t \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$

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$\Rightarrow f(t) > a - \frac{1}{n}, \forall\ n \in \mathbb{N}$

$\Rightarrow f(t) \geq a, \forall\ a \in \mathbb{R}$ ~~$\Rightarrow t \in$~~

$\Rightarrow t \in \{x \in E : f(x) \geq a\}$

$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\} \subseteq \{x : f(x) \geq a\}$ → (b)

from (a) and (b)

$\Rightarrow \{x \in E : f(x) \geq a\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$

$\because \{x \in E : f(x) > a - \frac{1}{n}\}$ is measurable.

$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$ is also measurable.

$\forall\ n \in \mathbb{N}$

$\Rightarrow \{x \in E : f(x) \geq a\}$ is also measurable.

Topic #68

(ii) $\Rightarrow$ (iii) $\{x \in E : f(x) \geq a\} \in M \Rightarrow \{x \in E : f(x) < a\} \in M$.

$\{x \in E : f(x) < a\} = \{x \in E : f(x) \geq a\}^c \quad \forall\ a \in \mathbb{R}$

$\because \{x \in E : f(x) \geq a\} \in M \Rightarrow \{x \in E : f(x) \geq a\}^c \in M$

$\Rightarrow \{x \in E : f(x) < a\} \in M$ i.e. measurable.

Topic #69

(iii) $\Rightarrow$ (iv) $\underbrace{\{x \in E : f(x) < a\}}_{\in M} \Rightarrow \{x \in E : f(x) < a\} \in M$.

ple show that $\{x \in E : f(x) \leq a\} =$

$= \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\}$

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Let $y \in \{x \in E : f(x) \leq a\}$ ——— (a)

$\Rightarrow f(y) \leq a$, $\forall\ a \in R$

$\Rightarrow f(y) < a + \frac{1}{n}$, $\forall\ n \in N$

$\Rightarrow y \in \{x \in E : f(x) < a + \frac{1}{n}\}$, $\forall\ n \in N$

$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\}$ ——— (B)

From (a) and (B) $\Rightarrow \{x \in E : f(x) \leq a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\}$

Let $z \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\}$

$\Rightarrow z \in \{x \in E : f(x) < a + \frac{1}{n}\} \Rightarrow f(z) < a + \frac{1}{n}, \forall n \in N$

$\Rightarrow f(z) \leq a\ \forall\ a \in R \Rightarrow z \in \{x \in E : f(x) \leq a\}$

$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\} \subseteq \{x \in E : f(x) \leq a\}$

$\therefore \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\} = \{x \in E : f(x) \leq a\}$

Given that $\{x \in E : f(x) < a + \frac{1}{n}\} \in M$.

$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\} \in M$

$\Rightarrow \{x \in E : f(x) \leq a\} \in M$. i.e is measurable.

Topic # 70

(iv) $\Rightarrow$ (i) ; $\{x \in E : f(x) \leq a\} \in M$, $\Rightarrow$ "f" is measurable.

$f : E \to R \cup \{+\infty\}$, $\{x \in E : f(x) < a\} \in M$,

$\{x \in E : f(x) \leq a\} \in M$, $\forall\ a \in R$

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$= \{x \in E : f(x) \leq a\}^c = \{x \in E : f(x) > a\} \in M$

$\Rightarrow$ "f" is measurable.

Topic # 71

Theorem: Let $\phi \neq E \in M$, $f : E \to R \cup \{\pm\infty\}$, then $\{x \in E : f(x) = a\} \in M$, $\forall a \in R$.

Proof: Case I: $a \in R$

$\Rightarrow \{x \in E : f(x) = a\} = \{x \in E : f(x) \geq a\} \cap \{x \in E : f(x) \leq a\}$

$\therefore \{x \in E : f(x) \geq a\}$ and $\{x \in E : f(x) \leq a\} \in M$.

$\Rightarrow \{x \in E : f(x) \geq a\} \cap \{x \in E : f(x) \leq a\} \in M$.

$\Rightarrow \{x \in E : f(x) = a\} \in M$ (measurable)

Case II: If $a = \infty$, Then $\{x \in E : f(x) = \infty\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) > n\}, \forall n \in N$

Let $y \in \{x \in E : f(x) = \infty\} \Rightarrow y \in \{x \in E : f(x) > n\}$

$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) > n\}, \forall n \in N$

$\Rightarrow x \in E : f(x) = \infty \subset \bigcap_{n=1}^{\infty} \{x \in E : f(x) > n\}$ ——— (a)

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Now let $t \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) > n\}, \forall n \in \mathbb{N}$

$\Rightarrow t \in \{x \in E : f(x) > n\}, \forall n \in \mathbb{N}$

$\Rightarrow f(t) > n, \forall n \in \mathbb{N}$

$\Rightarrow t \in \{x \in E : f(x) = \infty\}$

$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E : f(x) > n\} \subseteq \{x \in E : f(x) = \infty\} \longrightarrow (b)$

From (a) and (b)

So, $\{x \in E; f(x) = \infty\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) > n\}$

Case III: If $a = -\infty$, Then $\{x \in E : f(x) = -\infty\} =$

$$\bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}$$

Let $z \in \{x \in E : f(x) = -\infty\} \Rightarrow z \in \{x \in E : f(x) < -n\}$

$\Rightarrow z \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}, \forall n \in \mathbb{N}$

$\Rightarrow \{x \in E : f(x) = -\infty\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\} \longrightarrow (a)$

Now let $s \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}, \forall n \in \mathbb{N}$

$\Rightarrow s \in \{x \in E : f(x) < -n\}, \forall n \in \mathbb{N}$

$\Rightarrow f(s) < -n, \forall n \in \mathbb{N}$

$\Rightarrow s \in \{x \in E : f(x) = -\infty\}$

$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\} \subseteq \{x \in E : f(x) = -\infty\} \longrightarrow (b)$

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From (a) and (b)

So, $\{x \in E : f(x) = -\infty\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}$

Topic # 72

Positive anel Negative parts of Extented Real Valued Function:

$$|f(x)| = \begin{cases} +f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

$$f^+ = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ 0 & \text{if } f(x) < 0 \end{cases}$$

$$= \max\{f(x), 0\} = f \vee 0$$

$$f^- = \begin{cases} 0 & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

$$= \max\{-f(x), 0\} = -f \vee 0$$

where $f : A \to \mathbb{R} \cup \{\pm\infty\}$, $A \neq \phi$, $\forall x \in A$

Topic # 73

Theorem: Show that

(i) $f^+ = \frac{f + |f|}{2}$, (ii) $f^- = \frac{|f| - f}{2}$, (iii) $f = f^+ - f^-$

(iv) $|f| = f^+ + f^-$, (v) $\max(f, g) = \frac{(f+g) + |f-g|}{2}$

(vi) $\min(f, g) = \frac{(f+g) - |f-g|}{2}$

proof:

(i) If $f(x) \geq 0 \Rightarrow f^+(x) = f(x) \Rightarrow f^+ = f$

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$$|f(x)| = |f|(x) = f(x)$$

$$\therefore \frac{f+|f|}{2}(x) = \frac{f(x)+|f|(x)}{2} = \frac{f(x)+f(x)}{2} = f(x) = f^+(x)$$

If $f(x) < 0 \Rightarrow f^+(x) = 0$, $|f|(x) = |f(x)| = -f(x)$

$$\therefore \frac{f+|f|}{2}(x) = \frac{f(x)+|f|(x)}{2} = \frac{f(x)-f(x)}{2} = 0 = f^+(x)$$

$$\Rightarrow \frac{f+|f|}{2} = f^+ \quad \text{(proved)}$$

(ii) If $f(x) > 0 \Rightarrow f^-(x) = 0$, $|f(x)| = |f|(x) = +f(x)$

$$\therefore \frac{|f|-f}{2}(x) = \frac{|f|(x)-f(x)}{2} = \frac{+f(x)-f(x)}{2} = 0 = f^-(x)$$

If $f(x) < 0 \Rightarrow f^-(x) = -f(x)$, $|f(x)| = |f|(x) = -f(x)$

$$\therefore \frac{|f|-f}{2}(x) = \frac{|f|(x)-f(x)}{2} = \frac{-f(x)-f(x)}{2} = -f(x) = f^-(x)$$

$$\Rightarrow \frac{|f|-f}{2} = f^- \quad \text{(proved)}$$

(iii) $f = f^+ - f^-$

If $f(x) \geq 0 \Rightarrow f^+(x) = f(x)$, $f^-(x) = 0$

$$\therefore (f^+ - f^-)(x) = f^+(x) - f^-(x) = f(x) - 0 = f(x) = f$$

$$\Rightarrow f^+ - f^- = f \quad \text{(proved)}.$$

(iv) $|f| = f^+ + f^-$

If $f(x) \geq 0 \Rightarrow f^+(x) = f(x)$, $f^-(x) = 0$

$$\therefore (f^+ + f^-)(x) = f^+(x) + f^-(x) = f(x) + 0 = f(x) = |f(x)|$$

$$\Rightarrow f^+ + f^- = |f| \quad \text{(proved)}.$$

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(V) $\max(f,g) = \dfrac{(f+g)+|f-g|}{2}$

$$|f-g| = \begin{cases} +(f-g), & \text{if } \geq 0 \\ -(f-g), & \text{if } < 0 \end{cases}$$

$$|f-g|(x) = |f(x)-g(x)| = \begin{cases} f(x)-g(x), & f(x) \geq g(x) \\ g(x)-f(x), & f(x) < g(x) \end{cases}$$

Case I: If $f(x) \geq g(x) \Rightarrow f(x)-g(x) \geq 0$

$\Rightarrow |f(x)-g(x)| = f(x)-g(x)$

$\therefore \dfrac{(f+g)+|f-g|}{2}(x) = \dfrac{f(x)+g(x)+|f(x)-g(x)|}{2}$

$= \dfrac{f(x)+g(x)+f(x)-g(x)}{2} = f(x)$

Note: $f(x) \geq g(x)$ then $\max(f,g) = f(x)$

$\Rightarrow \dfrac{(f+g)+|f-g|}{2} = \max(f,g)$

Case II: If $f(x) < g(x) \Rightarrow f(x)-g(x) < 0$

$\Rightarrow |f(x)-g(x)| = -(f(x)-g(x)) = g(x)-f(x)$

and if $f(x) < g(x)$ then $\max(f,g) = g(x)$

$\therefore \dfrac{(f+g)+|f-g|}{2}(x) = \dfrac{f(x)+g(x)+|f(x)-g(x)|}{2}$

$= \dfrac{f(x)+g(x)+g(x)-f(x)}{2} = g(x)$

$\Rightarrow \dfrac{(f+g)+|f-g|}{2} = \max(f,g)$ (proved)

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(vi) $\min(f,g) = \dfrac{(f+g)-|f-g|}{2}$

$$|f-g|(x) = |f(x)-g(x)| = \begin{cases} f(x)-g(x) & ; f(x) \geq g(x) \\ g(x)-f(x) & ; f(x) < g(x) \end{cases}$$

Case I: If $f(x) \geq g(x) \Rightarrow f(x)-g(x) \geq 0$

$\Rightarrow |f(x)-g(x)| = f(x)-g(x)$ and

If $f(x) \geq g(x) \Rightarrow \min(f,g) = g(x)$

$\therefore \dfrac{(f+g)-|f-g|}{2}(x) = \dfrac{f(x)+g(x)-|f(x)-g(x)|}{2}$

$= \dfrac{f(x)+g(x)-(f(x)-g(x))}{2}$

$= \dfrac{f(x)+g(x)-f(x)+g(x)}{2} = g(x)$

$\Rightarrow \dfrac{(f+g)-|f-g|}{2} = \min(f,g)$

Case II: If $f(x) < g(x) \Rightarrow f(x)-g(x) < 0$

$\Rightarrow |f(x)-g(x)| = g(x)-f(x)$ and

If $f(x) < g(x) \Rightarrow \min(f,g) = f(x)$

$\therefore \dfrac{(f+g)-|f-g|}{2}(x) = \dfrac{f(x)+g(x)-|f(x)-g(x)|}{2}$

$= \dfrac{f(x)+g(x)-g(x)+f(x)}{2} = f(x)$

$\Rightarrow \dfrac{(f+g)-|f-g|}{2} = \min(f,g)$ (Proved).

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Topic # 74

Characteristic Function:

Let $\phi \neq X$ and $A \subseteq X$, then characteristic function $\chi_A : X \to R$ is defined and given by:

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

If $A = \phi \Rightarrow \chi_A(x) = 0 = 0(x)$, $\forall x \in X$ (zero function)

$\Rightarrow \chi_\phi = 0$

And if $A = X \Rightarrow \chi_A(x) = 1 = 1(x)$ $\forall x \in X$ (Identity function)

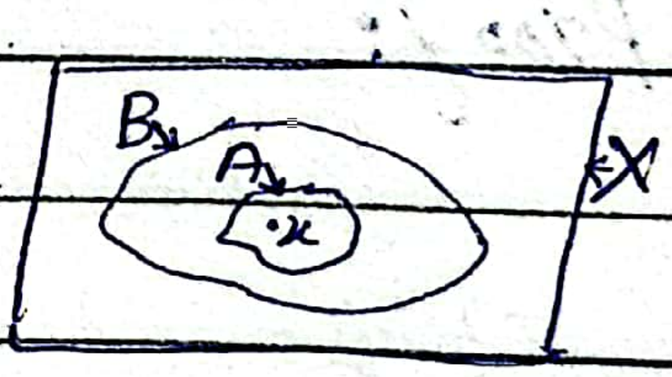
$\Rightarrow \chi_X = 1$

Topic # 75

Theorem: Prove that if $A \subseteq B$, then $\chi_A \leq \chi_B$.

Proof: Case I; If $x \in A$ and $A \subseteq B \Rightarrow x \in B$

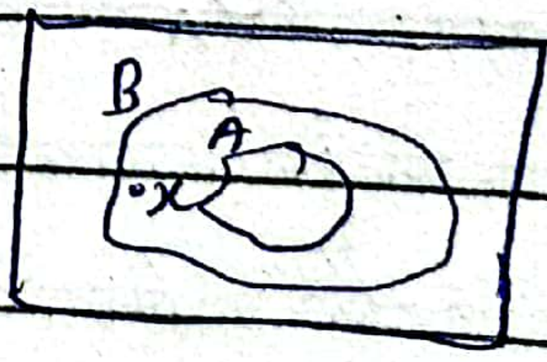
$\Rightarrow \chi_A = 1 = \chi_B \Rightarrow \chi_A \leq \chi_B$



Case II; If $x \in B$ but $x \notin A$

$\Rightarrow \chi_A = 0$ and $\chi_B = 1$

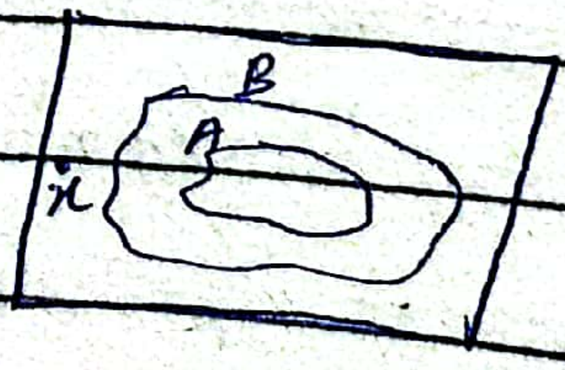
$\Rightarrow \chi_A \leq \chi_B$



Case III; If $x \notin A$ nor $x \notin B$

$\Rightarrow \chi_A = 0 = \chi_B \Rightarrow \chi_A \leq \chi_B$

(proved)



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Topic # 76

Theorem: Show that $\chi_{A\cap B} = \chi_A \cdot \chi_B$ 1

Proof: If $x \in A \cap B \Rightarrow x \in A$ and $x \in B$

$\Rightarrow \chi_{A\cap B} = 1 = \chi_A = \chi_B = 1 \cdot 1 = \chi_A \cdot \chi_B$

If $x \notin A\cap B \Rightarrow$ {
- $x \in A$ but $x \notin B \Rightarrow \chi_A = 1, \chi_B = 0$
- $x \in B$ but $x \notin A \Rightarrow \chi_B = 1, \chi_A = 0$
- $x \notin A$ and $x \notin B \Rightarrow \chi_A = \chi_B = 0$

$\therefore \chi_A \cdot \chi_B = 0 = O(x)$ }

$x \notin A\cap B \Rightarrow \chi_{A\cap B} = 0$

For all, $x \notin A\cap B$; $\chi_{A\cap B} = 0 = O(x) = \chi_A \cdot \chi_B$

$\chi_{A\cap B} = \chi_A \cdot \chi_B$ (proved)

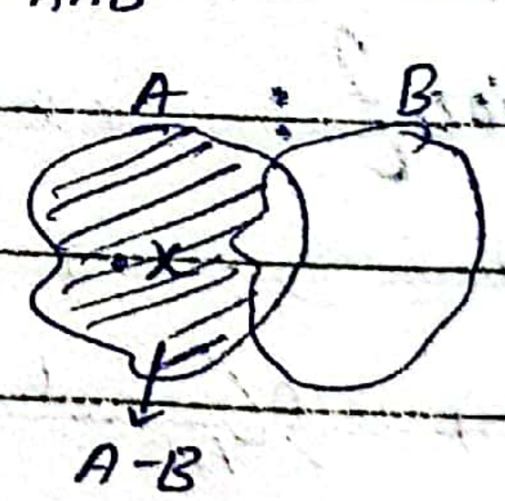
Topic # 77

Theorem: Show that $\chi_{A-B} = \chi_A - \chi_{A\cap B}$.

Proof: Case I: If $x \in A - B$

$\Rightarrow x \in A$ but $x \notin B, x \notin A\cap B$

$\Rightarrow \chi_{A-B} = 1$



$\chi_A - \chi_{A\cap B} = 1 - 0 = 1 = \chi_{A-B}$

Case II If $x \notin A - B \Rightarrow$ {
- $x \notin A, x \notin B, x \notin A\cap B \longrightarrow$ (a)
- $x \notin A, x \in B, x \notin A\cap B \longrightarrow$ (b)
- $x \in A, x \in B, x \in A\cap B \longrightarrow$ (c)

$\Rightarrow \chi_{A-B} = 0 - O(x)$

$\chi_A - \chi_{A\cap B} =$ {
- (a) $\Rightarrow 0 - 0 = 0$
- (b) $\Rightarrow 0 - 0 = 0$
- (c) $\Rightarrow 1 - 1 = 0$

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$\therefore$ For all cases; $\chi_{A-B} = \chi_A - \chi_{A\cap B}$ (proved)

Topic #78

Theorem: Show that $\chi_{A\times B} = \chi_A \cdot \chi_B$

Proof: $A\times B = \{(x,y) : x\in A, y\in B\}$

Case I:

$(x,y) \in A\times B \Rightarrow \begin{cases} x\in A \text{ and } x\in B \\ \Rightarrow \chi_A = 1 = \chi_B \end{cases}$

$\Rightarrow \chi_{A\times B}(x,y) = 1$

$\therefore \chi_{A\times B} = \chi_A \cdot \chi_B$

Case II: $\begin{cases} x\in A \text{ but } y\notin B \rightarrow (a) \\ x\notin A \text{ but } y\in B \rightarrow (b) \\ x\notin A \text{ and } y\notin B \rightarrow (c) \end{cases}$

If $(x,y) \notin A\times B \Rightarrow$

$\Rightarrow \chi_{A\times B}(x,y) = 0$

Also: $\chi_A \cdot \chi_B = \begin{cases} (a) \Rightarrow 1\cdot 0 = 0 \\ (b) \Rightarrow 0\cdot 1 = 0 \\ (c) \Rightarrow 0\cdot 0 = 0 \end{cases}$

In all cases: $\chi_{A\times B} = \chi_A \cdot \chi_B$ (proved)

Topic #79

Theorem: If $A = \bigcup_{n=1}^{\infty} A_n$ and $\{A_n\}_{n=1}^{\infty}$ is a sequence of pairwise disjoint subsets of X. Then show that $\chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$

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Case I:

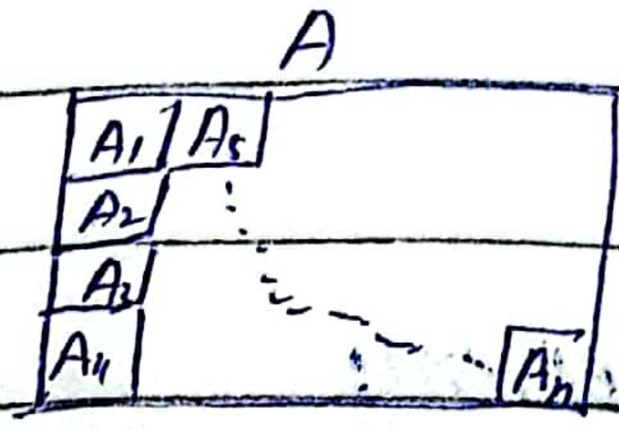
Proof: If $x \in A \Rightarrow x \in \bigcup_{n=1}^{\infty} A_n$

$\Rightarrow x \in A_k$ exactly for $n = k$

$x \notin \bigcup_{n=1}^{\infty} A_n - \{A_k\}$

$\Rightarrow \chi_{A_k}(x) = 1, \chi_{A_n}(x) = 0, \forall n \neq k$

$x \in A \Rightarrow \chi_A(x) = 1$



$\therefore \chi_A(x) = 1 = 0 + 0 + \cdots + \underset{n=k}{1} + 0 + 0 + \cdots$

$\chi_A(x) = 1 = \sum_{n=1}^{\infty} \chi_{A_n}(x) \Rightarrow \chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$

Case II: $x \notin A = \bigcup_{n=1}^{\infty} A_n \Rightarrow x \notin A_n, \forall n \in \mathbb{N}$

$\Rightarrow \chi_A(x) = 0$ $\left\{ \Rightarrow \chi_{A_n}(x) = 0, \forall n \in \mathbb{N} \right.$

$= 0 + 0 + 0 + \cdots + 0 + \cdots = \sum_{n=1}^{\infty} \chi_{A_n}(x)$

$\Rightarrow \chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$ (proved)

Topic # 80

Theorem: Let $\phi \neq A \subseteq X$ and for characteristic function $\chi_A : X \to \mathbb{R}$, χ_A is measurable $\Leftrightarrow$ "A" is measurable.

Proof: Suppose "χ_A" is measurable (function)

$\Rightarrow \forall a \in \mathbb{R}$ is set; $\{x : \chi_A(x) > a\}$ is measurable.

In particular for $a = 0$; $\{x : \chi_A(x) > 0\} = A$

By definition of χ_A.

$\therefore A \in M$ (measurable)

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Conversely, let $A \in M$ (measurable) and we prove "χ_A" is measurable.
In other words we want to prove the $\{x : \chi_A(x) \geq a\} \in M$, $\forall\ a \in R$

$$\{x : \chi_A(x) \geq a\} = \begin{cases} \phi & \text{if } a \geq 1 \\ A & \text{if } 0 \leq a < 1 \\ A \cup A^c & \text{if } a < 0 \end{cases}$$

(i) $a \geq 1$, $\chi_A(x) \geq 1 \Rightarrow$ there does not exist $x \in A$ such that $\chi_A(x) \geq 1$

(ii) $0 \leq a < 1$, $\chi_A(x) > 0.9$ (say)

(iii) $a < 0$ (say -2), $\chi_A(x) > -2$

In all cases; if $A \in M$ (measurable)

$\Rightarrow A^c \in M$ (measurable)

$\Rightarrow A \cup A^c \in M$ (measurable)

$\phi \in M$ (measurable)

$\Rightarrow \{x : \chi_A(x) \geq a\} \in M$ (measurable)

So "χ_A" is measurable. (Proved)

Topic # 81

Algebra of Measurable Functions:

If "f" and "g" are two extended real valued and measurable functions defined on same domain $\phi \neq D$ then (i) $f + k$, (ii) kf, (iii) $f \pm g$, (iv) fg, (v) $\frac{f}{g}, g \neq 0$, (vi) $|f|$, (vii) $f \vee g$ (viii) $f \wedge g$ are measurable functions.

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Topic # 82 & 83

Proof: is "$f+k$" is measurable, $k \in R$

$\because$ "f" is measurable $\Rightarrow \{x : f(x) > a\} \in M, \forall a \in R$

$\Rightarrow \{x : f(x) > a-k\} \in M$ (measurable)

$\Rightarrow \{x : f(x)+k > a\} \in M, \forall a \in R$

$\Rightarrow$ "$f+k$" is measurable.

83 (ii) "kf" is measurable, $\forall k \in R$ | $f \to kf$ Scaler composition

proof: $\because$ "f" is measurable.

$\Rightarrow \{x : f(x) > a\} \in M, \forall a \in R$

(a) If $k > 0$: $\because f(x) > a, \forall a \in R$

$\Rightarrow f(x) > \frac{a}{k} \Rightarrow kf(x) > a, \forall a \in R$

$\therefore \{x : kf(x) > a\} \in M \ \forall a \in R$

$\Rightarrow$ "kf" is measurable.

(b) If $k < 0$; $f(x) > a, \forall a \in R$ (given)

$\Rightarrow f(x) > \frac{a}{k} \Rightarrow kf(x) < a \quad (\because k < 0)$

$\Rightarrow \{x : kf(x) < a\} \in M$

$\Rightarrow$ "kf" is measurable.

(c) If $k = 0 \Rightarrow \{x : kf(x) > a\}, \forall a \in R$

$= \{x : 0 f(x) > a\} \ \because \forall a \in R$

$= \{x : 0 > a\}, \forall a \in R$ ... (1)

$\{x : 0 > a\} = \begin{cases} D & ; \ a < 0 \ (\because D : \exists x : f(x) > a) \\ \phi & ; \ a \geq 0 \end{cases}$

"D" is measurable (given) and ϕ is measurable.

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$\therefore \{x : |kf(x)| > a\}$ is measurable.

$\Rightarrow$ "kf" is measurable.

Topic # 84

(iii) "$f+g$" is measurable.

proof: We have to proof "$f+g$" is measurable.

$\Rightarrow \{x : (f+g)(x) > a\} \in M$ (To prove)

$\{x : (f+g)(x) > a\} = \{x : f(x) + g(x) > a\}$

$= \{x : f(x) > a - g(x)\}$, $\forall\, a \in R$

$\because \{f(x), a - g(x) \in R \Rightarrow \exists\, p \in Q$ such that

$f(x) > p > a - g(x)\}$

because "Q" is dense in R : $\bar{Q} = R$

$\therefore \{x : (f+g)(x) > a\} = \bigcap_{p \in Q} \{x : f(x) > p\} \cap \{x : a - g(x) < p\}$

i.e. our claim.

Let $y \in \{x : (f+g)(x) > a\} \Rightarrow y \in \{x : f(x) + g(x) > a\}$

$\Rightarrow f(y) + g(y) > a \Rightarrow f(y) > a - g(y)$

$\Rightarrow$ there exist($\exists$) $p \in Q$ such that $f(y) > p > a - g(y)$

$Q = R$

$\Rightarrow f(y) > p$ and $a - g(y) < p$

$\Rightarrow y \in \{x : f(x) > p\}$ and $y \in \{x : a - g(x) < p\}$

$\Rightarrow y \in \{x : f(x) > p\} \cap \{x : a - g(x) < p\}$

$\Rightarrow y \in \bigcup_{p \in Q} [\{x : f(x) > p\} \cap \{x : a - g(x) < p\}]$

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$\Rightarrow \{x : (f+g)(x) > a\} \subseteq \bigcup_{p \in \mathbb{Q}} \left[\{x : f(x) > p\} \cap \{x : a - g(x) < p\}\right]$ ——→ (A)

Now Let $t \in \bigcup_{p \in \mathbb{Q}} \left[\{x : f(x) > p\} \cap \{x : a - g(x) < p\}\right]$

$\Rightarrow t \in \{x : f(x) > p\} \cap \{x : a - g(x) < p\}$ for some $p \in \mathbb{Q}$

$\Rightarrow t \in \{x : f(x) > p\}$ and $t \in \{x : a - g(x) < p\}$

$\Rightarrow f(t) > p$ and $a - g(t) < p \Rightarrow p > a - g(t)$

$\Rightarrow f(t) > p > a - g(t) \Rightarrow f(t) > a - g(t)$

$\Rightarrow f(t) + g(t) > a \rightarrow (f+g)(t) > a$

$\Rightarrow t \in \{x : (f+g)(x) > a\}$

$\Rightarrow \bigcup_{p \in \mathbb{Q}} \left[\{x : f(x) > p\} \cap \{x : a - g(x) < p\}\right] \subseteq \{x : (f+g)(x) > a\}$ ——→ (B)

From (A) and (B) $\rightarrow \{x : (f+g)(x) > a\} =$

$$= \bigcup_{p \in \mathbb{Q}} \left[\{x : f(x) > p\} \cap \{x : a - g(x) < p\}\right]$$

$\therefore$ $\begin{cases} \text{"}f\text{" is measurable} \Rightarrow \{x : f(x) > p\} \in M \\ \text{"}g\text{" is measurable} \Rightarrow \{x : a - g(x) < p\} \in M \end{cases}$

$\Rightarrow \{x : f(x) > p\} \cap \{x : a - g(x) < p\} \in M$, for $p \in \mathbb{Q}$

$\Rightarrow \bigcup_{p \in \mathbb{Q}} \left[\{x : f(x) > p\} \cap \{x : a - g(x) < p\}\right] \in M$

$\Rightarrow \{x : (f+g)(x) > a\} \in M$ (measurable)

$\Rightarrow$ "$f+g$" is measurable function.

Now we have to prove "f-g" is measurable.

⇒ As g is measurable ⇒ (-1)g = -g is also M.

⇒ f + (-1)g = f - g is also measurable.

(proved)

Topic#85

(iv) "fg" is measurable.

proof: $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$

Firstly we prove "f^2" is measurable.

⇒ we need to prove $\{x : f^2(x) > a\} \in M, \forall a \in R$.

$y = x^2$ (Say function)

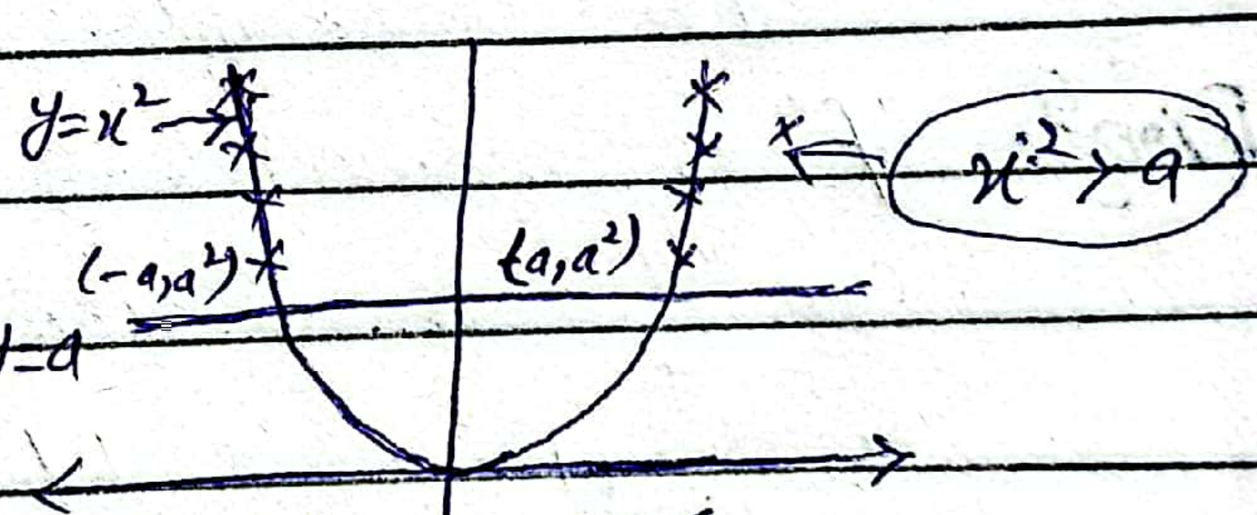
$y = a$ (line)

$\Rightarrow y^2 = a^2$

$y = x^2$

$a^2 = x^2$

$\Rightarrow x = \pm a$



Case 2: $f^2(x) > a$

$\Rightarrow f(x) > a$, $f(x) < -a$

$\{x : f^2(x) > a\} = \begin{cases} \{x : f(x) > \sqrt{a}\} \cup \{x : f(x) < -\sqrt{a}\}, & a \geq 0 \\ \text{All Domain} \cdot (D) & , a < 0 \end{cases}$

∴ "f" is measurable ⇒ $\{x : f(x) > \sqrt{a}\} \in M$

and $\{x : f(x) < -\sqrt{a}\} \in M$

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$\Rightarrow \{x: f(x) > \sqrt{q}\} \cup \{x: f(x) < -\sqrt{q}\} \in M$

and "Domain" is given to be measurable. i.e $D \in M$

$\therefore \{x: f^2(x) < q\} \in M \Rightarrow$ "f^2" is measurable.

$\because f, g$ is measurable $\Rightarrow (f \pm g)$ is also measurable.

$\Rightarrow (f \pm g)^2$ is measurable.

$\Rightarrow \frac{1}{4}[(f+g)^2 - (f-g)^2] = fg$ is also measurable.

($\because Kf$ is also measurable) $\forall K \in R$.

Topic # 86

prove that if "f" is measurable function, then "$\frac{1}{f}$" is also measurable.

Proof: "f" is measurable $\Rightarrow \{x: f(x) > a\} \in M, \forall a \in R$.

To prove "$\frac{1}{f}$" is measurable, we show that

$\{x: \frac{1}{f}(x) > a\} \in M, \forall a \in R$.

$$\frac{1}{f}(x) = \frac{1}{f(x)} > a$$

$$\left\{x: \frac{1}{f}(x) > a\right\} = \begin{cases} \{x: f(x) < \frac{1}{a}\} \cap \{x: f(x) > 0\}; & a > 0 \\ \{x \in D: f(x) > 0\}; & a = 0 \\ \{x: f(x) > 0\} \cup [\{x: f(x) < \frac{1}{a}\} \cap \{x: f(x) < a\}] & ; a < 0 \\ \phi \end{cases}$$

All the sets on R.H.S are measurable.

$\Rightarrow \{x: \frac{1}{f}(x) > a\}$ is also measurable, $\forall a \in R$.

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$\Rightarrow$ $\frac{1}{f}$ is measurable.

(v) $\frac{f}{g}$, $g \neq 0$, is measurable.

proof: $\frac{f}{g}$ is also measurable because $\frac{f}{g} = \frac{1}{g} \cdot f$ is measurable being product of two measurable functions.

Topic # 87

(vi) $|f|$ is measurable.

proof: $\because$ "f" is measurable $\Rightarrow \{x : f(x) > a\} \in M, \forall a \in R$

To prove $|f|$ is measurable, we need to show:

$\{x : |f|(x) > a\} \in M, \forall a \in R$

$$\{x : |f|(x) > a\} = \{x : |f(x)| > a\}$$

$$= \begin{cases} \underbrace{\{x : f(x) > a\}}_{\in M} \cup \underbrace{\{x : f(x) < -a\}}_{\in M}, & \forall a > 0 \\ \{x : |f|(x) > a\} = \\ D \in M, & a < 0 \end{cases}$$

In all cases on R.H.S, we get a measurable set. $\Rightarrow \{x : |f|(x) > a\} \in M$ (measurable), $\forall a \in R$

$\Rightarrow |f|$ is also measurable.

(vii) $f \vee g = \max(f, g)$ is measurable.

proof: $\because f \vee g = \max(f, g) = \frac{1}{2}[(f+g) + |f-g|]$

So $f \vee g$ is measurable because $(f+g)$ and $|f-g|$ is measurable and "kf" is measurable, $\forall\ k \in \mathbb{R}$.

(viii) $f \wedge g = \min(f, g)$ is measurable.

proof: $\because f \wedge g = \min(f, g) = \frac{1}{2}[(f+g) - |f-g|]$

So $f \wedge g$ is measurable because $(f+g)$ and $|f-g|$ is measurable and "kf" is measurable, $\forall\ k \in \mathbb{R}$.

Topic # 88

Restriction of a function:

Let $f: X \to \mathbb{Y}$ be a function and $A \subset X$, Then $g: A \to \mathbb{Y}$ defined as $g(x) = f(x)$, $x \in A$ is called the restriction of "f" to $A \subset X$ and given by $f_A = g$.

Theorem: Let "f" be an extended real valued measurable function on "D". Let $A \subset D$, Then restriction of "f" to "A" is also measurable and $A \in \mathcal{M}$.

Proof: $\because$ "f" is measurable $\Rightarrow \{x \in D : f(x) > a\} \in \mathcal{M}$. $\forall\ a \in \mathbb{R}$

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For any $a \in \mathbb{R}$, To prove "f_A" is measurable we have $\{x \in A : f_A(x) > a\} \in M$

Note: $\{x \in A : f_A(x) > a\} = \{x \in A : f(x) > a\} \cap A$

On R.H.S both the sets are measurable.

$\Rightarrow \{x \in A : f_A(x) > a\} \in M$

$\therefore$ Intersection of two measurable sets is measurable.

$\Rightarrow$ "f_A" is measurable.

Topic # 89

Theorem: An extended real valued function defined on a set of measure zero is measurable.

Proof: Let $\phi \neq E \subset \mathbb{R}$, such that $m^*(E) = 0$ and $f : E \to \mathbb{R} \cup \{\pm\infty\}$, To prove "$f$" is measurable, we need to show that:

$\{x \in E : f(x) > a\} \in M$, $\forall\ a \in \mathbb{R}$.

$\therefore \{x \in E : f(x) > a\} \subset E$

$\Rightarrow m^*\{x \in E : f(x) > a\} \leq m^*(E)$

$\because$ By monotonicity of m^*.

$\Rightarrow m^*\{x \in E : f(x) > a\} \leq 0$ ..... (given)

But by non-negativity of "m^*"

$\Rightarrow m^*\{x \in E : f(x) > a\} \geq 0$

$\Rightarrow m^*\{x \in E : f(x) > a\} = 0$; $\forall\ a \in \mathbb{R}$.

$\Rightarrow \{x \in E : f(x) > a\} \in M \; ; \forall \; a \in \mathbb{R}$

So "f" is measurable.

Topic # 90

Theorem: Let $\{f_n\}$ be a sequence of extended real valued measurable function on "D", then.

(i) $\sup_{n \in \mathbb{N}} f_n$, $\inf_{n \in \mathbb{N}} f_n$ are measurable.

(ii) $\max_{1 \le i \le n} (f_i)$, $\min_{1 \le i \le n} (f_i)$ are measurable.

(iii) $\overline{\lim_{n \in \mathbb{N}}} f_n$, $\underline{\lim_{n \in \mathbb{N}}} f_n$ are measurable.

(iv) If $\lim_{n \to \infty} f_n = f$, Then "f" is measurable.

Proof: Let $f = \inf_{n \in \mathbb{N}} f_n$ and To prove f is measurable.

Then we have to shows $\{x : f(x) > a\} \in M, \forall a \in \mathbb{R}$

We claim : $\{x : f(x) > a\} = \bigcup_{n=1}^{\infty} \{x : f_n(x) > a\}$

Let $y \in \{x : f(x) > a\} \Rightarrow f(y) > a$

$\because f = \inf f_n \Rightarrow f(y) \le f_n(y), \forall \; n \in \mathbb{N}$

$\because f(y) > a \Rightarrow a < f(y) \le f_n(y), \forall \; n \in \mathbb{N}$

$\Rightarrow f_n(y) > a \; \forall \; n \in \mathbb{N} \Rightarrow y \in \{x : f_n(x) > a\} \; \forall \; n \in \mathbb{N}$

$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x : f_n(x) > a\}$

$\Rightarrow \{x : f(x) > a\} \subseteq \bigcap_{n=1}^{\infty} \{x : f_n(x) > a\} \longrightarrow (A)$

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Now let $t \in \bigcap_{n=1}^{\infty} \{x : f_n(x) > a\}$

$\Rightarrow t \in \{x : f_n(x) > a\}, \forall a \in R, \forall n \in N$

$\Rightarrow f_n(t) > a, \forall n \in N$

$\Rightarrow a < f_n(t), \forall n \in N$

$\Rightarrow$ "a" is lower bound $\{f_1(t), f_2(t), \cdots\}$

But $f = \inf_{n \in N} f_n$ is also its lower bound.

$\Rightarrow a < \inf_{n \in N} f_n, \forall n \in N$

$\Rightarrow$ ~~$f_n(t) > a$~~ $\Rightarrow \inf_{n \in N} f_n > a \Rightarrow f(t) > a$

$f(t) > a \Rightarrow t \in \{x : f(t) > a\} \longrightarrow (A)$

$\Rightarrow \bigcap_{n=1}^{\infty} \{x : f_n(x) > a\} \subset \{x : f(x) > a\} \longrightarrow (B)$

From (A) and (B) $\{x : f(x) > a\} = \bigcap_{n=1}^{\infty} \{x : f_n(x) > a\}$

So $f = \inf_{n \in N} f_n$ is measurable.

Now let $h = \sup_{n \in N} f_n$

... The set $\{x : h(x) > a\} = \bigcup_{n=1}^{\infty} \{x : f_n(x) > a\}$ is measurable. So $h = \sup_{n \in N} f_n$ is measurable.

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Topic# 9

$\max_{1\le i\le n} f_i$, $\min_{1\le i\le n} f_i$ are measurable.

$\{f_n\}_{n=1}^{\infty}$ is seq of measurable functions.

Proof: Let $f = \max_{1\le i\le n} f_i$, To prove "f" is measurable, we need to show that $\{x : f(x) > a\} \in M$ and we claim

$$\{x : f(x) > a\} = \bigcup_{k=1}^{n} \{x : f_k(x) > a\} \quad \forall a \in \mathbb{R}.$$

let $y \in \{x : f(x) > a\} \Rightarrow f(y) > a$

$\therefore f(x) = \max_{1\le i\le n} f_i$ So there exist $J = \{1, 2, \dots, n\}$

$\Rightarrow f(x) = f_j(x) \Rightarrow f(y) = f_j(y) > a$

$\Rightarrow y \in \{x : f_j(x) > a\}$, for some $j = \{1, 2, \dots, n\}$

Hence $y \in \bigcup_{i=1}^{n} \{x : f_i(x) > a\}$

Now let $y \in \bigcup_{i=1}^{n} \{x : f_i(x) > a\}$

Then $y \in \{x : f_k(x) > a\}$ for some $k = \{1, 2, \dots, n\}$

$\Rightarrow f_k(y) > a$, By definition $f_i(y) \le \max f_i(y) = f(y)$ $\forall i = 1, 2, \dots, n.$

and $f_k(y) \le f(y) \Rightarrow h(y) > a$

$\Rightarrow y \in \{x : f(x) > a\}$.

The countable union of measurable sets is measurable. So $f = \max_{1\le i\le n} f_i$ is measurable.

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$\overline{\lim_{n\in\mathbb{N}}}$ → Limit superior

$\underline{\lim_{n\in\mathbb{N}}}$ → Limit inferior

Let $g = \min_{1\le i\le n} f_i$, To prove "g" is measurable we need to show that $\{x : g(x) > a\} \in M \quad \forall a \in \mathbb{R}$

and we claim $\{x : g(x) > a\} = \bigcap_{i=1}^{n} \{x : f_i(x) > a\}$

The countable intersection of measurable sets is measurable. So $g = \min_{1\le i\le n} f_i$ is measurable.

Topic #92

Show that $\overline{\lim_{n\in\mathbb{N}}} f_n, \underline{\lim_{n\in\mathbb{N}}} f_n$ are measurables.

Proof:

$$\overline{\lim_{n\in\mathbb{N}}} f_n = \inf_{n\in\mathbb{N}} \left(\sup_{k\ge n} f_k \right) \rightarrow (9)$$

$$\sup_{k\ge n} f_k = \sup\{f_k, f_{k+1}, f_{k+2}, \dots\}, \quad k = 1, 2, 3, \dots$$

$$= \begin{cases} \sup\{f_1, f_2, f_3, \dots\} = F_1 \text{ is measurable being "sup" of measurable functions.} \\ \sup\{f_2, f_3, f_4, \dots\} = F_2, \dots \end{cases}$$

$\Rightarrow \{F_n\}_{n=1}^{\infty}$ is a sequence of measurable functions

$\Rightarrow \inf_{n\in\mathbb{N}} \{F_n\}_{n=1}^{\infty}$ is also measurable

$\Rightarrow \overline{\lim_{n\in\mathbb{N}}} f_n = \inf_{n\in\mathbb{N}} \{f_n\} = \inf_{n\in\mathbb{N}} \left(\sup_{k\ge n} f_k \right)$ is also measurable

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Similarly $\underline{\lim}_{n\in\mathbb{N}} f_n = \sup_{n\in\mathbb{N}} (\inf_{k\geq n} f)$ is measurable.

Topic # 93

If $\lim_{n\to\infty} f_n = f$, then "f" is measurable.

Proof: $\because \lim_{n\to\infty} f_n = \overline{\lim}_{n\in\mathbb{N}} f_n = \underline{\lim}_{n\in\mathbb{N}} f_n = f$ $\{\{f_n\}_{n=1}^{\infty}$ ↓ Limit (given)

But $\overline{\lim}_{n\in\mathbb{N}} f_n$ and $\underline{\lim}_{n\in\mathbb{N}} f_n$ are measurable.

$\Rightarrow$ "f" is also measurable.

MTH644 - Measure Theory Handouts

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